# DATA2DASH

## AI-Powered Multi-Agent Research Platform

**Course:** Mobile Computing
**Course Supervisor:** Dr. Ahmed Hussein

**Team**

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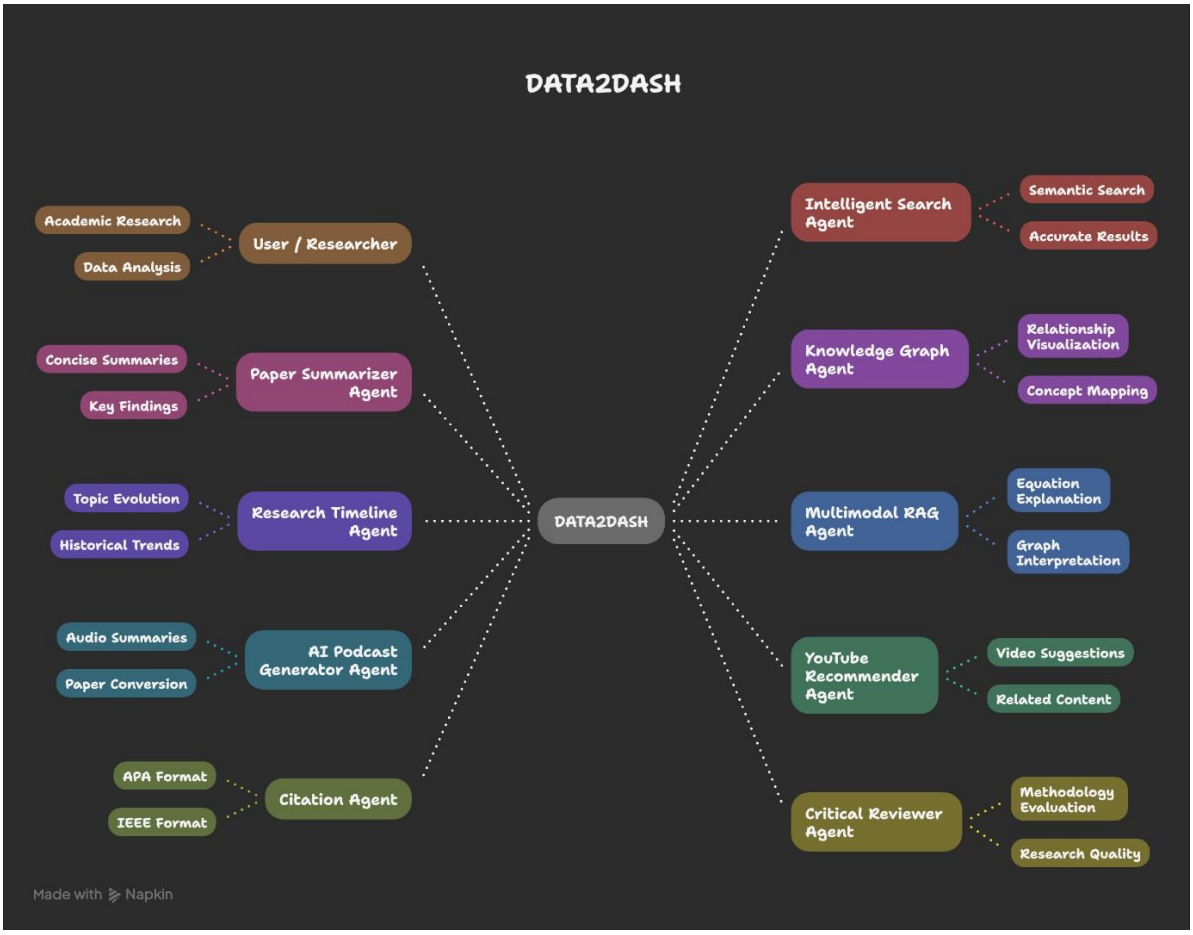
# Introduction

DATA2DASH is an AI-powered multi-agent research platform designed to help students, researchers, and professionals search, understand, summarize, analyze, and organize academic papers in one unified workspace.

Researchers usually depend on multiple disconnected tools such as Google Scholar for discovery, ChatGPT for explanation, PDF readers for reading, and Word documents for writing notes. This tool fragmentation increases cognitive load and slows down the literature review process. DATA2DASH solves this issue by combining intelligent semantic search, paper summarization, multimodal RAG, knowledge graph visualization, research timeline, podcast generation, YouTube recommendations, citation generation, and critical review in one platform.

The main goal of DATA2DASH is to transform academic research from a manual and time-consuming process into a faster, smarter, and more organized workflow. Instead of only searching and reading papers, users can move toward understanding, analyzing, and discovering insights more efficiently.



# Section 01 — UX Phase

## 1.1 UX Stage I — Research

The research phase aimed to understand the difficulties users face during academic research and literature review. A user survey was conducted to collect insights about users' current research habits, tools, pain points, preferred features, and interface preferences.

The survey collected **57 responses**. The majority of respondents were students, representing **74%** of the sample. This indicates that the primary target users are students and early researchers who need support in searching, understanding, and organizing academic papers. The survey also showed that many users are comfortable using AI tools, which supports the idea of building an AI-powered research assistant.


## 1.2 Survey Questions

The survey included questions covering the following areas:

1. What is your current role?
2. How often do you work on literature review or academic research?
3. What tools do you currently use to search for papers?
4. What are the biggest challenges you face while reading or analyzing papers?
5. How comfortable are you with using AI tools?
6. Which features would be most useful in an academic research platform?
7. How much time do you usually spend on literature review per week?
8. What interface theme do you prefer: light mode, dark mode, system default, or toggle option?
9. Do you prefer reading, watching videos, or listening to audio summaries while learning?
10. What would make the research process easier for you?

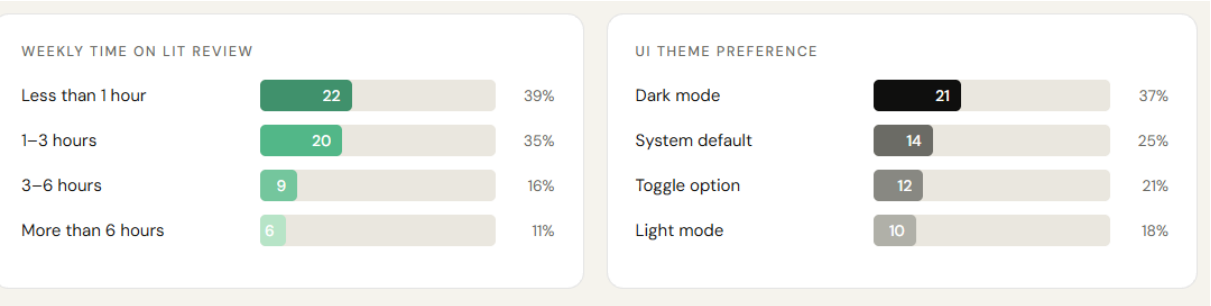
## 1.3 Survey Analysis

The survey results revealed that most users face several repeated problems during academic research. The most common pain points were information overload, difficulty understanding mathematical notation, difficulty finding research gaps, and the need to find relevant papers quickly.

The results showed that **53%** of respondents struggle with too many papers, while another **53%** struggle with hard mathematical notation. Around **47%** of users reported difficulty finding research gaps, and **28%** needed a faster way to find relevant papers. In addition, **23%** of respondents mentioned the need for audio or multitasking support.

The survey also showed that users currently depend on multiple tools. **G1%** use ChatGPT or AI tools, **56%** use Google Scholar, **1G%** use ResearchGate, **18%** use ArXiv, and **7%** use Semantic Scholar. This confirms that users already use digital and AI tools, but their workflow is still fragmented across many platforms.

Feature priority results showed that users valued intelligent search the most, with an average rating of **4.07/5**, followed by chat with PDF at **4.00/5**, summary generation at **3.88/5**, citation generator at **3.77/5**, knowledge graph at **3.67/5**, research timeline at **3.61/5**, YouTube recommender at **3.54/5**, and AI podcast at **3.28/5**.




## 1.4 Research Conclusions

Based on the survey results, the following conclusions were identified:

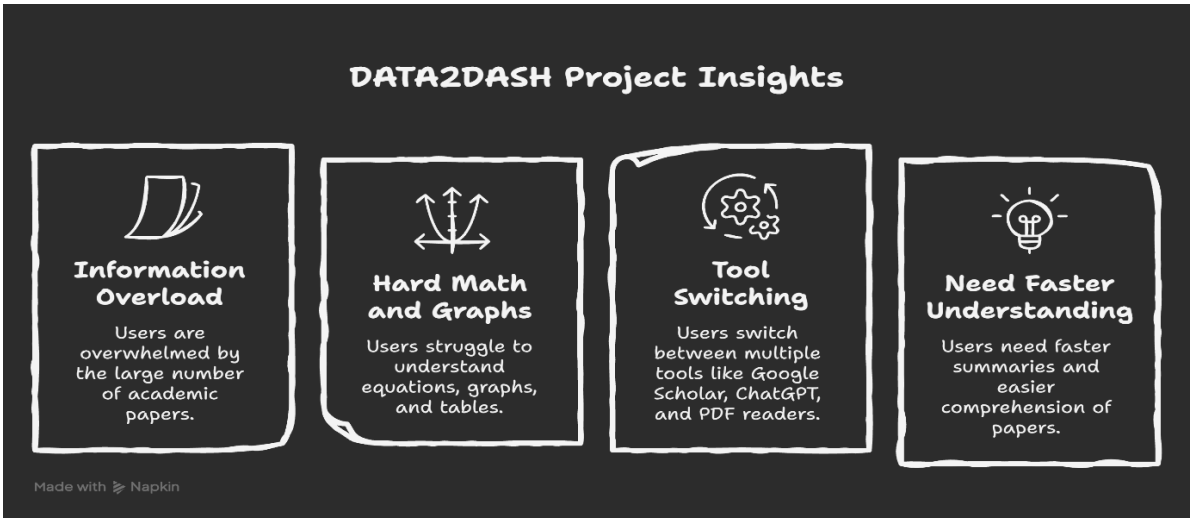
First, students represent the main target audience for DATA2DASH, as they formed the largest percentage of respondents. This means the platform should be simple, guided, and supportive for users who may not have advanced research experience.

Second, users suffer from information overload because they need to review many academic papers in a short time. Therefore, intelligent search and summarization are essential features.

Third, mathematical notation, graphs, and tables make papers harder to understand. This supports the need for a multimodal RAG feature that can explain equations, figures, and tables in a simpler way.

Fourth, users already rely heavily on ChatGPT and other AI tools, but they still use Google Scholar, PDF readers, and other platforms separately. This confirms the need for an all-in-one platform that reduces tool-switching.

Finally, users showed interest in features such as citation generation, knowledge graphs, research timelines, audio learning, and YouTube recommendations. These features can improve the research experience and make learning more flexible.



## Pain points and Gains

| Pain Points | Expected Gains |
|---|---|
| Information overload due to too many papers | Find relevant papers faster using intelligent search |
| Complex mathematics, graphs, and tables | Understand difficult content through multimodal RAG |
| Hard to identify research gaps | Discover relationships and gaps using knowledge graph and research timeline |
| Tool fragmentation and switching between platforms | Use one integrated platform for the full research workflow |
| No audio support for learning on the go | Listen to AI podcast summaries anytime |
| No quick summary for papers | Get concise summaries before reading the full paper |
| Manual citation formatting | Generate citations automatically in APA/IEEE |
| No visual/video support for difficult topics | Get YouTube recommendations for visual explanation |

## 1.5 Empathy Map

The empathy map summarizes what the primary users think, feel, say, and do while conducting academic research.

**Thinks:**
Users think that there are too many papers to follow and that finding the right papers quickly is difficult. They also feel that mathematical notation in academic papers slows down their understanding.

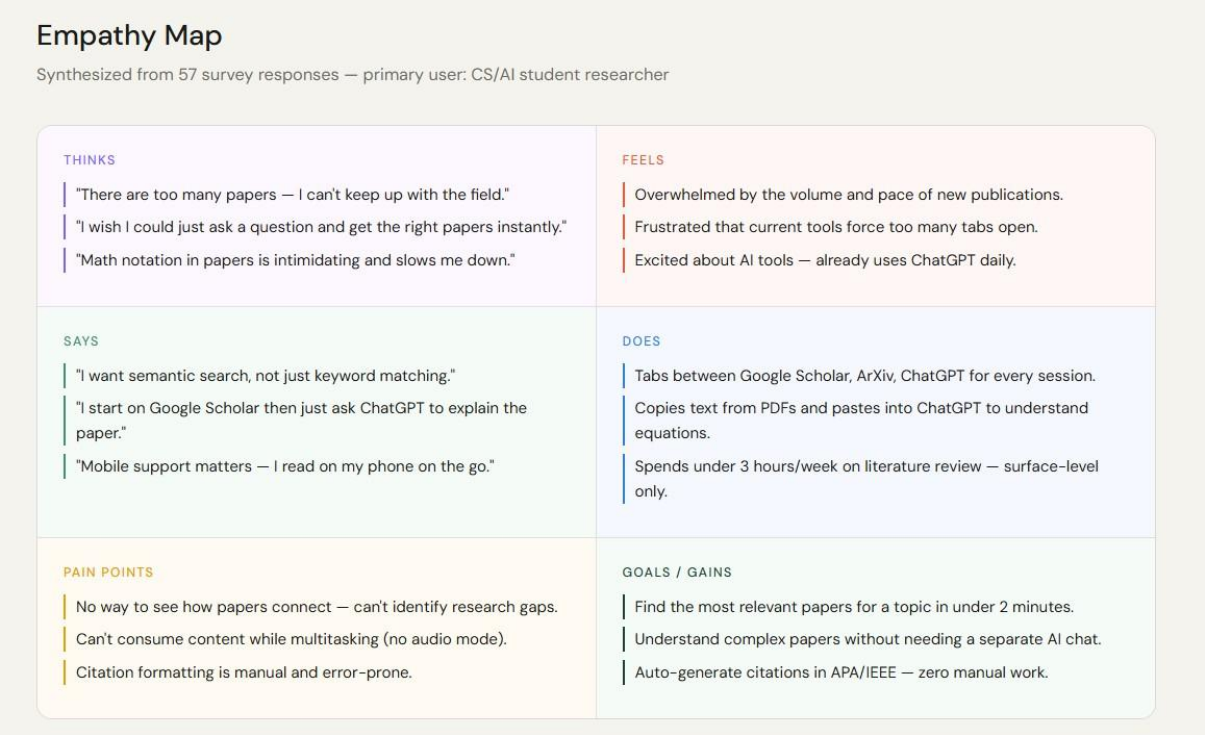
**Feels:**
Users feel overwhelmed by the large number of publications and frustrated because they have to use many tools at the same time. However, they are also excited about AI tools because they already use them in their daily learning process.

**Says:**
Users want semantic search instead of simple keyword matching. They also want a tool that can explain papers, summarize content, and support mobile or flexible learning.

**Does:**
Users switch between Google Scholar, ArXiv, ChatGPT, PDF readers, and writing tools. They often copy text from papers into AI tools to understand difficult sections.

**Pain Points:**
Users cannot easily see how papers connect, they cannot identify research gaps quickly, they lack audio learning support, and citation formatting is still manual.

**Gains:**
Users want to find papers quickly, understand complex papers easily, and generate citations automatically.




## 1.6 Primary Persona

**Name:** Ahmed Tarek
**Age:** 22
**Role:** Computer Engineering Student
**Location:** Cairo, Egypt
**University:** Helwan University

Ahmed is a fourth-year computer engineering student interested in AI and cybersecurity. He frequently needs to read academic papers for coursework, research, and

graduation-related topics. He is comfortable using AI tools and often uses ChatGPT to explain difficult concepts.

Ahmed's main problem is that academic research takes too much time. He switches between Google Scholar, ArXiv, ChatGPT, PDF readers, and note-taking tools during one research session. He also struggles with mathematical notation, long papers, and identifying how different papers are connected.

Ahmed needs a platform that helps him search for relevant papers, summarize them, explain difficult equations, generate citations, and visualize relationships between research topics. DATA2DASH supports Ahmed by providing intelligent search, chat with PDF, citation generation, knowledge graph visualization, and AI podcast summaries.


## 1.7 Secondary Persona

**Name:** Sara Mahmoud
**Role:** Machine Learning Engineer
**User Type:** Industry Professional

Sara uses academic research to stay updated with new AI and machine learning methods. She does not have enough time to read full papers deeply, so she needs fast summaries, highly relevant papers, and quick insight extraction. DATA2DASH helps her by providing search, summarization, knowledge graph, and critical review features.

## 1.8 Wireframes and Sitemap

The following figures show the low-fidelity wireframes created for the Data2Dash AI application. These wireframes were used to plan the main screens, user flow, navigation sidebar, and the layout of the core features before implementing the final UI. The wireframes cover authentication, chat, search, upload, document viewing, and citation-related screens.

**Sign In Screen**

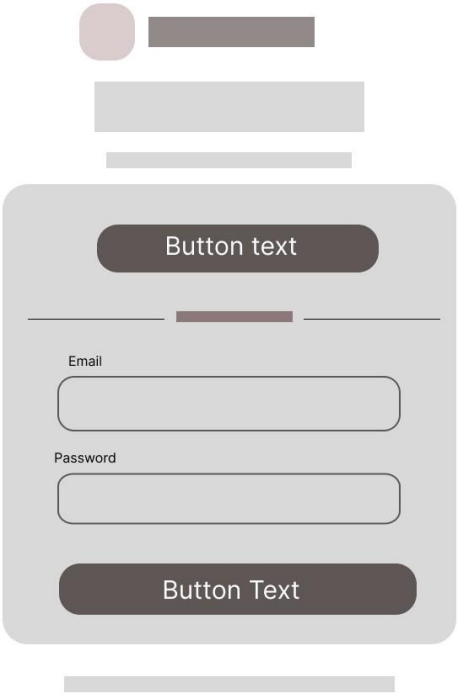



**Figure 1.8: Low-fidelity wireframe for the sign-in screen.**

This wireframe shows the sign-in screen where existing users can enter their email and password to access the application. It includes a simple authentication form and a main action button to continue to the system.

**Create Account Screen**

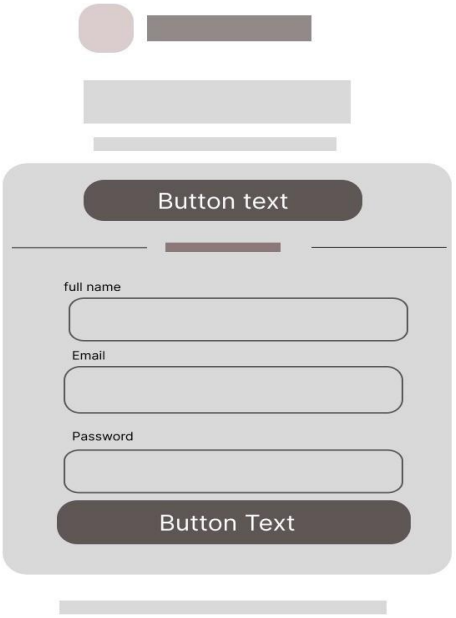



**Figure 1.G: Low-fidelity wireframe for the create account screen.**

This wireframe shows the registration screen where new users can create an account by entering their full name, email, and password. This screen represents the first step in the user onboarding flow.

**Main Chat Screen**

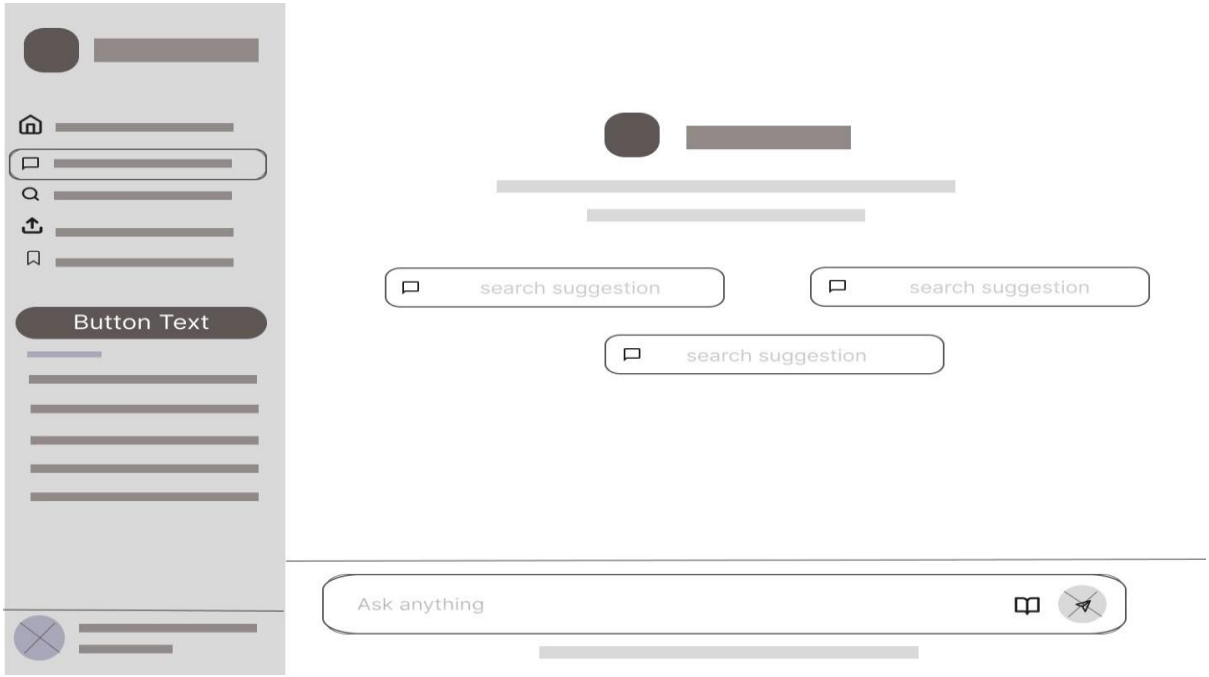


**Figure 1.10: Low-fidelity wireframe for the main chat screen.**

This wireframe represents the main chat interface of the application. It includes the navigation sidebar, suggested research questions, and a message input area where users can ask questions and receive AI-assisted responses.

**Search Screen**


**Figure 1.11: Low-fidelity wireframe for the search screen.**

This wireframe shows the search feature where users can search for academic topics, papers, or stored research content. It includes a search input, suggested results, filter options, and a results area.

### Upload Screen

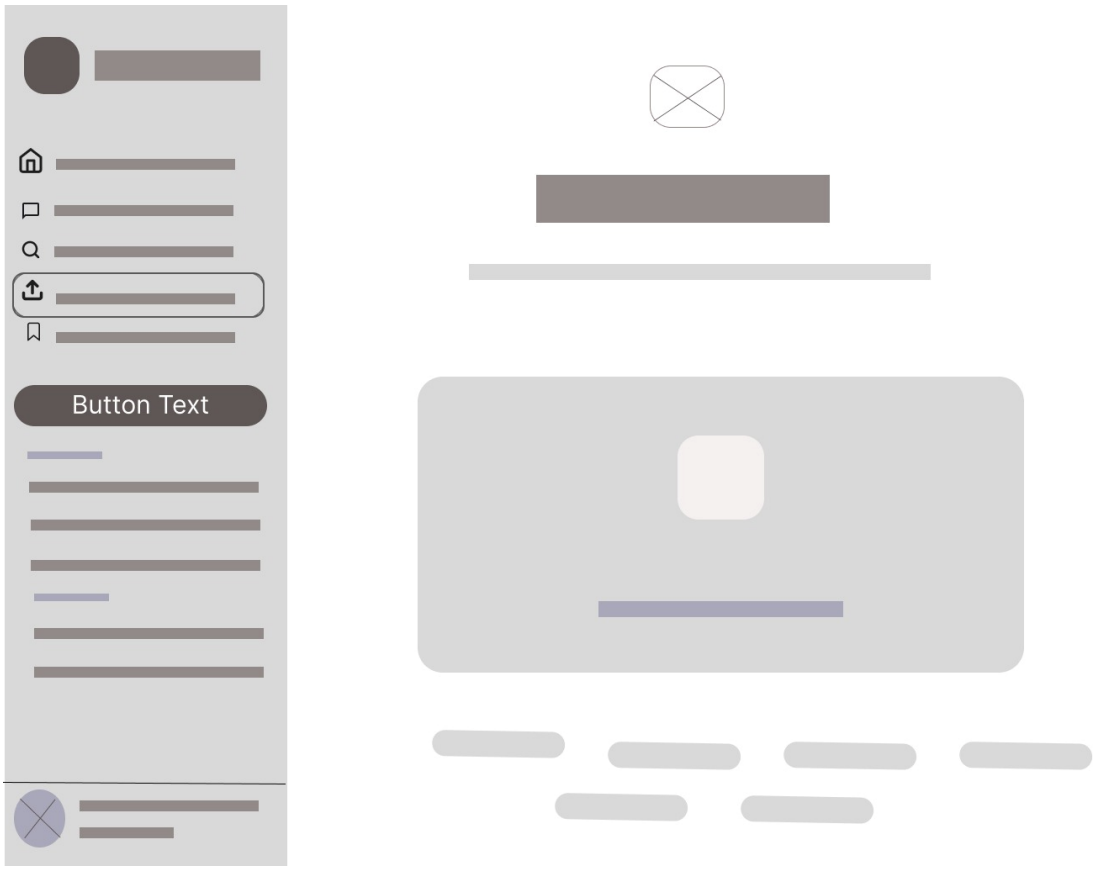



**Figure 1.12: Low-fidelity wireframe for the upload screen.**

This wireframe shows the file upload screen where users can upload research papers or PDF documents. The central upload area helps users easily add files to the system for later analysis and interaction.

### Document Viewer and Chat Screen

**Figure 1.13: Low-fidelity wireframe for the document viewer and chat screen.**

This wireframe shows the document viewer layout after a file is uploaded. The left side displays the selected document, while the right side provides a chat panel where users can ask questions about the uploaded file content.

**paper Details / Summary Modal**


Figure 1.14: Low-fidelity wireframe for the paper details and summary modal.

This wireframe represents a detailed paper view or summary modal. It provides an area for displaying paper information such as title, abstract, and related details, while the right side can display generated content, summaries, or analysis results.

**Citation / Document Reading Screen**


Figure 1.15: Low-fidelity wireframe for the citation and document reading screen.

This wireframe shows the citation and document reading screen. It includes a central document reading area, a navigation sidebar, and a right-side panel for related citation information, summaries, or supporting research content.

# Section 02: Frontend

### 2.1 Frontend Overview

The frontend of **DATA2DASH** is designed as a web-based AI research dashboard that helps users search, understand, summarize, visualize, cite, and listen to academic research content in one place. The main goal of the interface is to reduce tool-switching between Google Scholar, ChatGPT, PDF readers, and document editors by providing a unified research platform. The project presentation describes DATA2DASH as an **AI-powered multi-agent research platform** that moves users from "searching and reading" toward "understanding and discovering."

The frontend is implemented using **React 18**, **TypeScript**, and **Vite**, with **Tailwind CSS** for styling and responsive layout. The GitHub full-stack repository also lists React Router, Zustand, Framer Motion, Headless UI, Lucide React, and i18next as part of the frontend stack.

The interface includes main screens such as authentication, home/dashboard, academic search, upload/PDF interaction, citation generation, and AI agent features. The frontend connects the user with different AI agents such as the Intelligent Search Agent, Summarizer, Citation Agent, Knowledge Graph Agent, Multimodal RAG Agent, Critical Reviewer, Podcast/Audio Agent, and the updated **Video Agent**.

### 2.2 Frontend Technologies Used

| Technology | Version / Type | Usage in DATA2DASH |
|---|---|---|
| **React** | React 18 | Building the main web user interface |
| **TypeScript** | TypeScript 5 | Adding type safety and improving code reliability |
| **Vite** | Vite build tool | Running and building the frontend project |
| **Tailwind CSS** | CSS framework | Styling pages, cards, buttons, layouts, and responsiveness |
| **React Router DOM** | Routing library | Navigation between pages such as Home, Search, Upload, and Citation |
| **Zustand** | State management | Managing frontend application state |
| **Framer Motion** | Animation library | Adding smooth UI animations and transitions |
| **Headless UI** | UI component library | Creating accessible interface components |
| **Lucide React** | Icon library | Displaying icons for search, upload, citation, audio, video, and agents |
| **i18next / react-i18next** | Internationalization | Supporting future multilingual interface options |

### 2.3 Frontend Folder Structure

https://github.com/Data2Dash/Data2Dash-frontend/tree/main/src/components

### 2.4 Design Colors and Typography

| Type | Color Name | Hex Code | Usage |
|---|---|---|---|
| **Primary** | Brand Black | #0F0F0E | Navigation bar, main buttons, logo, dark sections |
| **Secondary** | Forest Green | #1B4332 | Active states, selected elements, important highlights |
| **Accent** | Accent Green | #40916C | Icons, highlights, progress bars, active UI elements |
| **Background** | Cream | #F5F3ED | Main page background |
| **Secondary Background** | Warm Gray | #EAE7DF | Secondary buttons, cards, dividers |
| **Neutral** | White | #FFFFFF | Cards, panels, document areas |
| **Text** | Muted Text | #6B6B65 | Subtitles, captions, metadata, secondary labels |

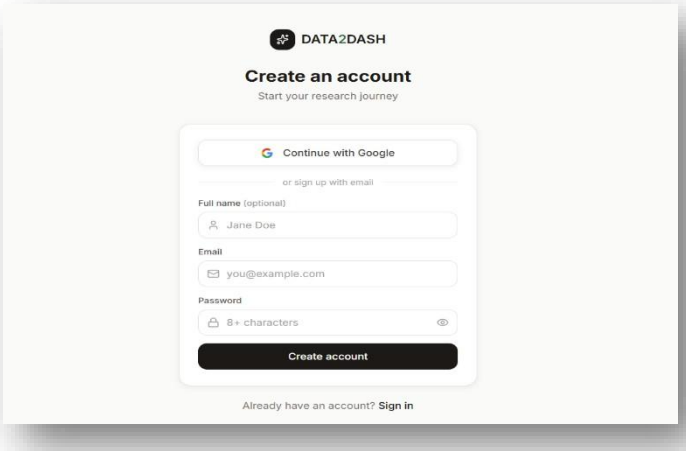
The typography used in the UX design is based mainly on **DM Sans**, which is used for headings, body text, buttons, labels, and captions. **DM Mono** is used for technical details such as metadata, color codes, and small system labels.

| Font | Usage |
|---|---|
| **DM Sans** | Headings, paragraphs, labels, buttons, navigation |
| **DM Mono** | Technical labels, metadata, hex codes, small system text |

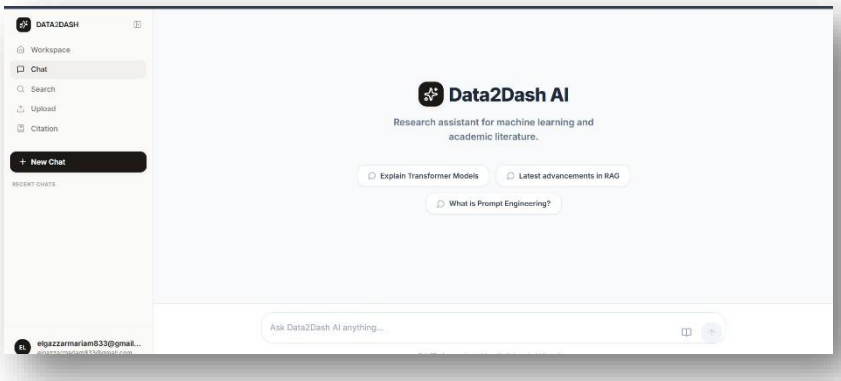
### 2.5 Main Frontend Screens

The frontend contains multiple screens that cover the main user workflow. The UX design file includes wireframes for core screens such as **Login**, **Chat**, **Upload / PDF Viewer**, **Search**, and **Citation**

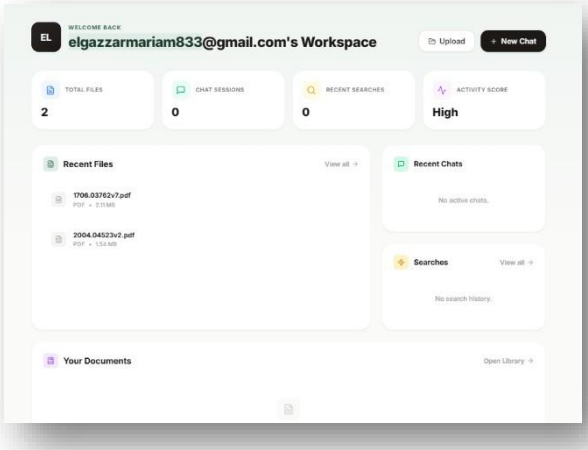
1. **Authentication Screen**
   The authentication screen includes both Sign in and Sign up interfaces. It allows users to create an account, sign in, continue with Google, and use email/password authentication.
   **Sign in and Sign up screenshots.**



2. **Sidebar Navigation**
   The application includes a fixed left sidebar that gives users access to the main sections of the platform: Workspace, Chat, Search, Upload, and Citation.
   **Workspace, Chat, Search, Upload, and Citation sidebar screenshots.**

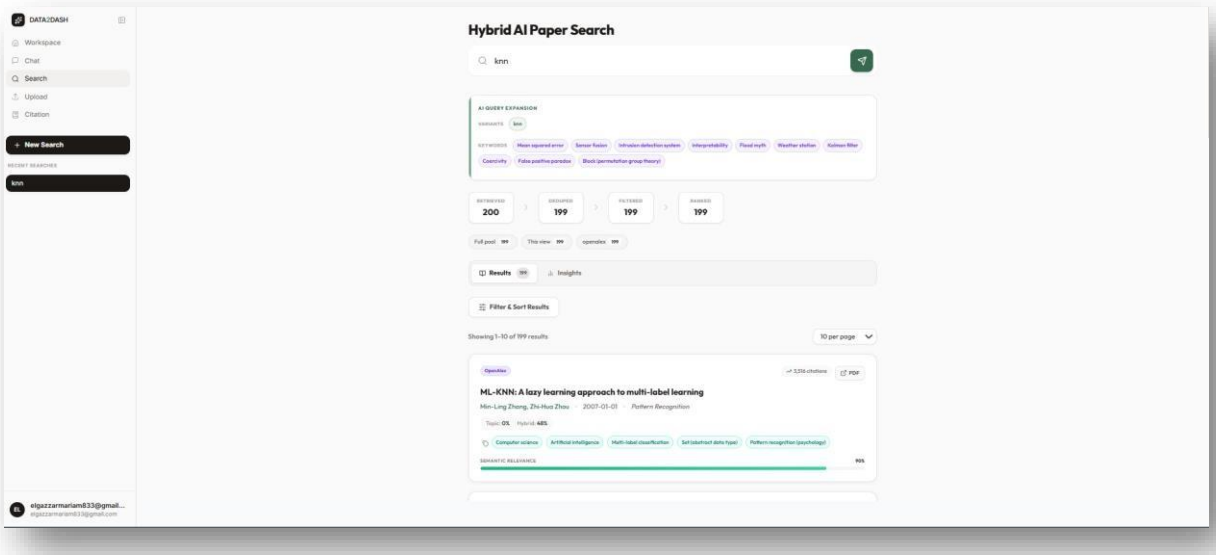



3. **Workspace Dashboard**
   The workspace dashboard acts as the user's main home page. It shows total files, chat sessions, recent searches, activity score, recent files, recent chats, and search history**.**
   **Workspace dashboard screenshot.**

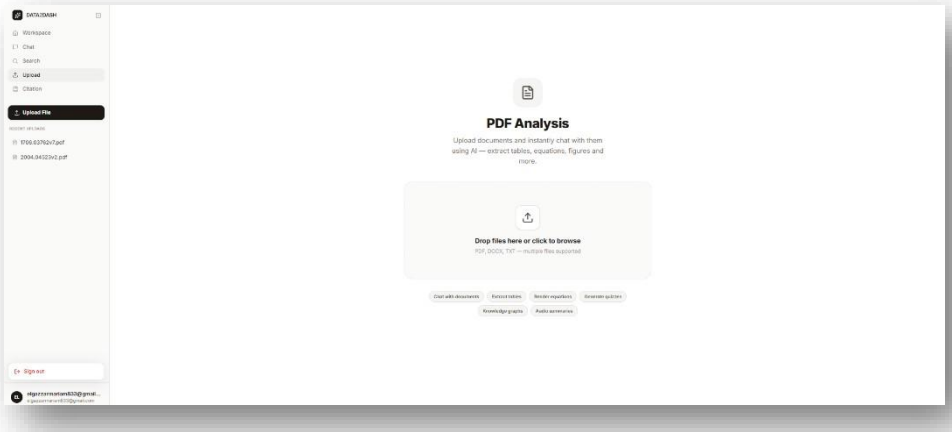



4. **Chat Screen**
   The chat screen is titled Data2Dash AI and allows users to ask research questions. It also includes suggested prompts such as explaining transformer models, latest advancements in RAG, and prompt engineering**.**
   **Data2Dash AI chat screenshot.**

5. **Hybrid AI Paper Search Screen**
   The search screen provides an academic search interface with a search input and suggested topics such as RAG, Transformers, GAN, BERT, and diffusion models. It supports AI-assisted paper search.
   **Hybrid AI Paper Search screenshot.**

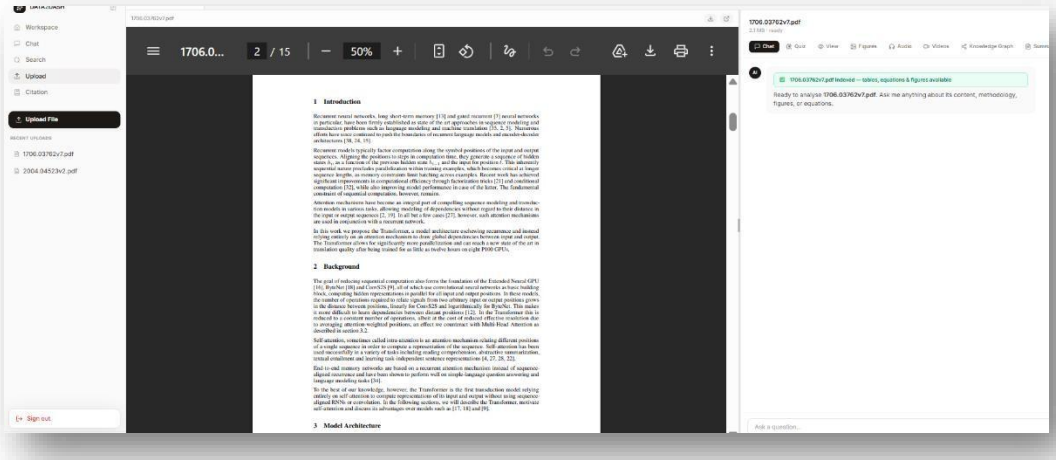



6. **Upload Screen**
   The upload screen is titled PDF Analysis. It allows users to upload documents such as PDF, DOCX, and TXT files for AI-based analysis.
   Evidence from screenshots: PDF Analysis upload screenshot.
   **Add the upload page screenshot.**

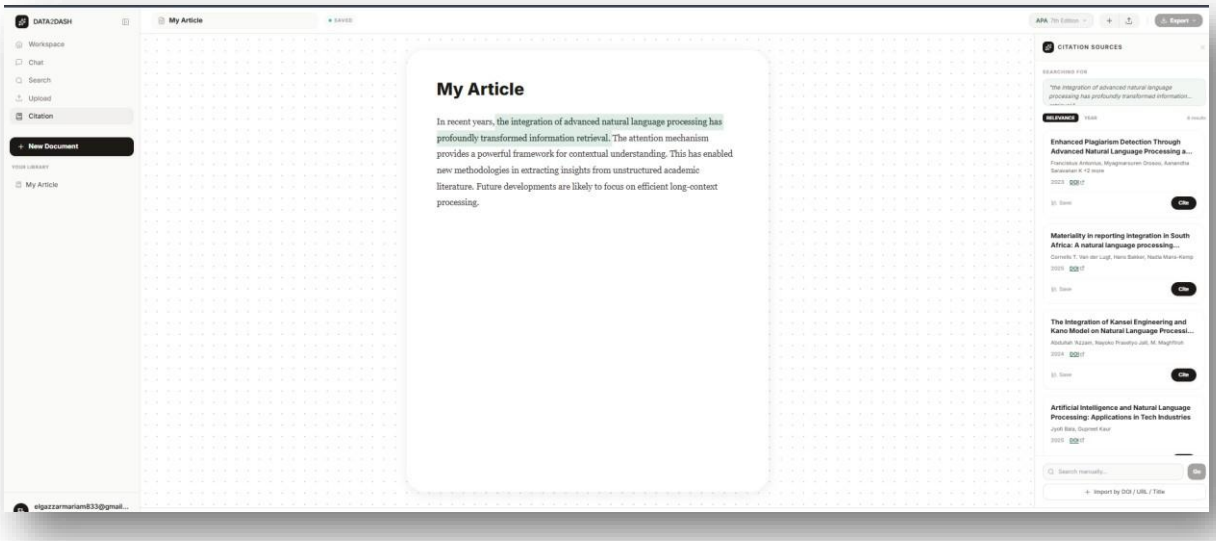


7. **Document Analysis Workspace**
   After uploading a PDF, the document workspace shows the PDF viewer and multiple AI document tools. These include Chat, Quiz, View, Figures, Audio, Videos, Knowledge Graph, and Summarize.

   **Uploaded PDF analysis screenshots.**



8. **Citation Editor**
   The citation screen provides an article editor where users can write academic text, select a citation format such as APA 7th Edition, export content, and view word count and character count**.**
   **My Article citation editor screenshot.**



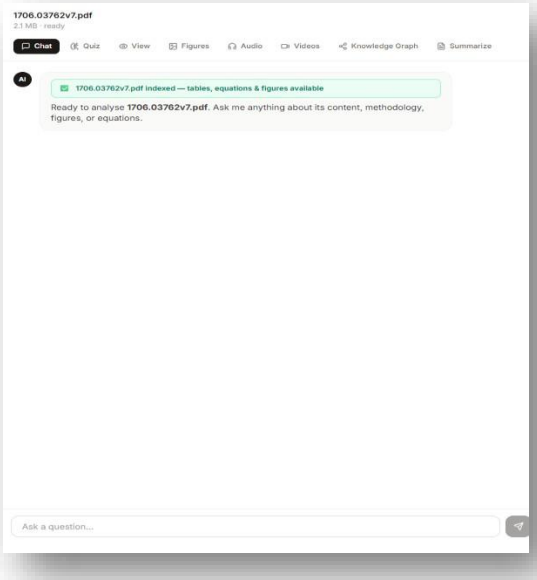
9. **Knowledge Graph Processing**

   The document analysis workspace includes a Knowledge Graph feature. The screenshot shows the system extracting knowledge graph entities from an uploaded document. This supports the project's goal of helping users understand relationships inside research papers.

   **Knowledge graph extraction screenshot.**

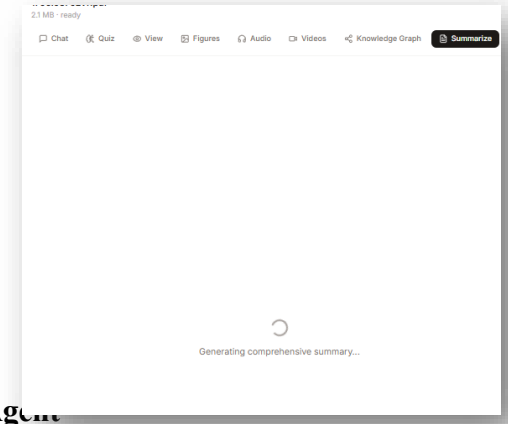

10. **Multimodal RAG / PDF Analysis Agent**

    The upload and PDF analysis screen allows users to upload documents and interact with them using AI. The document workspace includes tabs for Chat, Quiz, View, Figures, Audio, Videos, Knowledge Graph, and Summarize.



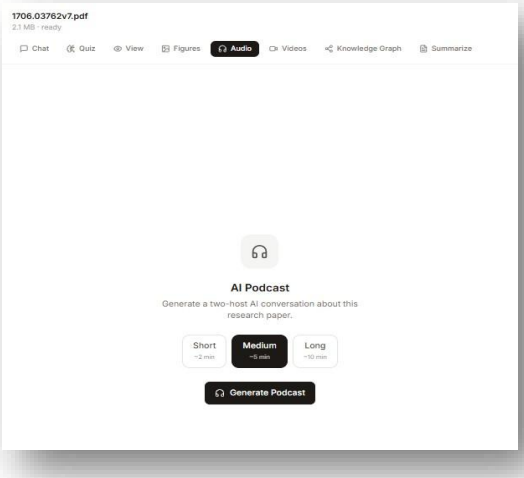
**11. Paper Summarizer Agent**

The summarizer feature helps users generate a short summary of an uploaded research paper. In the document workspace, the Summarize tab shows that summarization is part of the document analysis tools.



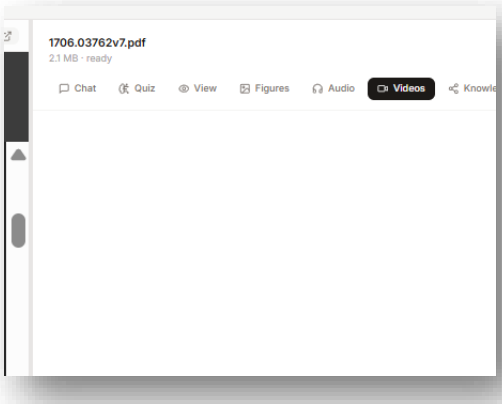

**12. Audio / Podcast Agent**

The audio feature allows users to listen to paper explanations or audio summaries. In the document workspace, the Audio tab shows that audio support is included as part of the platform.



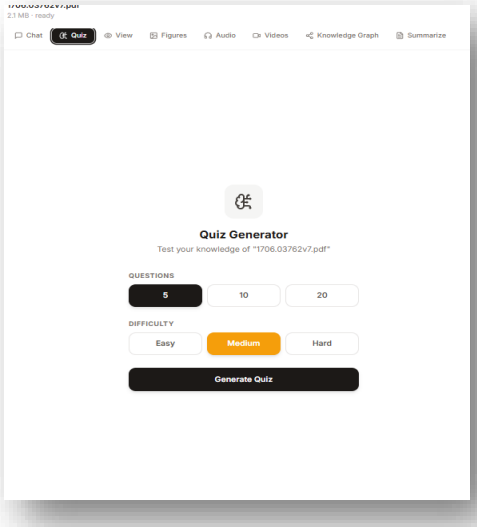

**13. Video Agent**

The Video Agent replaces the original YouTube Video Recommender idea. Instead of only recommending videos, it generates a simple presentation about a selected topic and explains it using audio. In the document workspace, the Videos tab shows that video support is included in the fronte



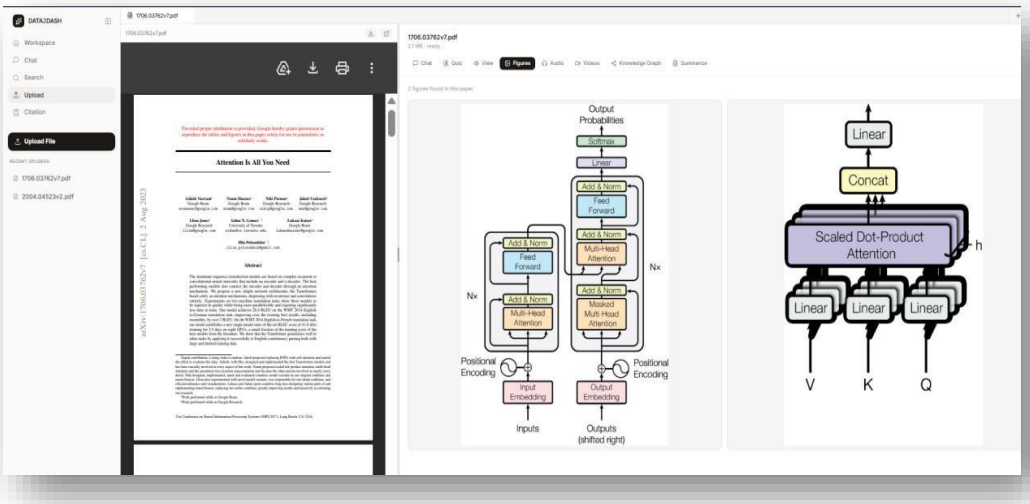
## 14. Quiz Agent

The document workspace includes a Quiz tab, which can be used to generate questions based on the uploaded document. This helps users test their understanding of the paper content.

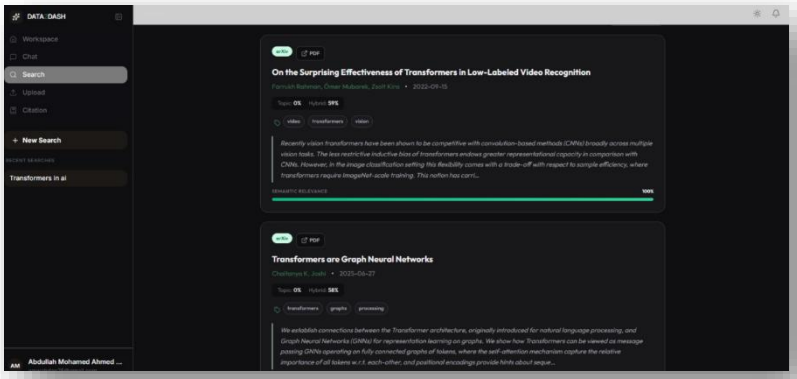


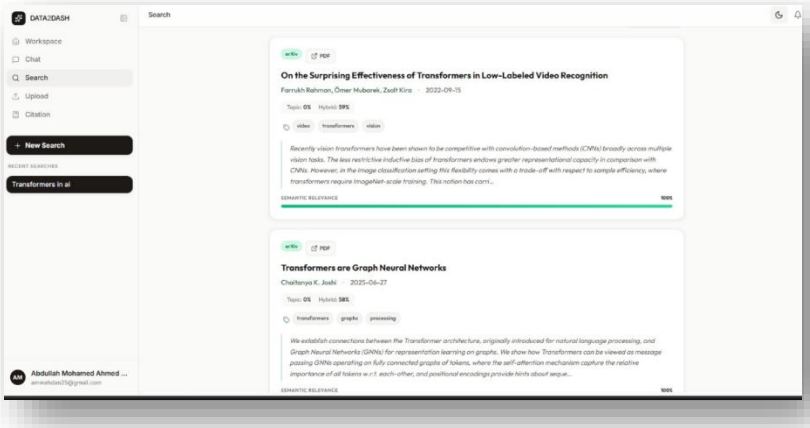

## 15. Figure Extractor Agent

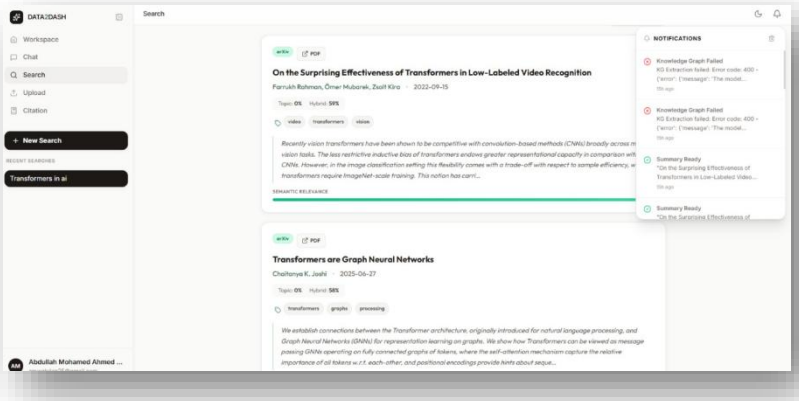
The Figure Extractor Agent is included in the document analysis workspace through the Figures tab. This feature is used to extract or display figures from uploaded research papers, helping users understand diagrams, charts, and visual content without manually searching through the full PDF. It supports the DATA2DASH goal of making academic papers easier to understand through text, visual, and audio-based tool



**16.** Dark/Light Mode, and Notifications

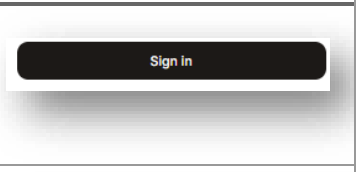
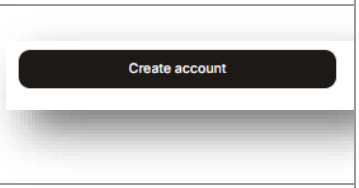
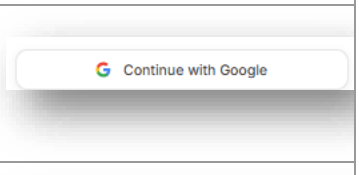
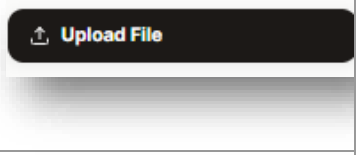
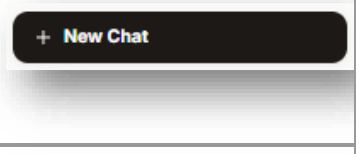
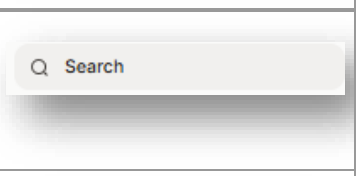
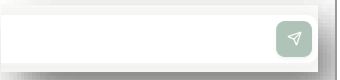
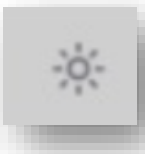
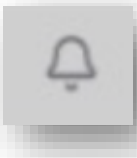



## 2.6 Custom Buttons

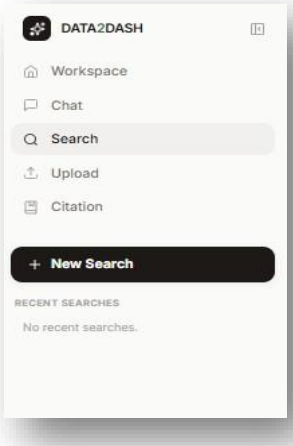
DATA2DASH uses reusable button components to keep the interface consistent. Primary actions such as signing in, searching, generating, and exporting use strong visual emphasis, while secondary actions use lighter colors. Icons are used beside buttons where needed to make actions easier to understand.

| Component Name | Style | Used In | Description | Screenshot to Add |
|---|---|---|---|---|
| **SignInButton** | Black filled button with rounded corners | Authentication screen | Used to sign in to an existing account |   |
| **CreateAccountButton** | Wide black primary button | Sign-up form | Submits the create account form |   |
| **GoogleButton** | White outlined button with Google icon | Authentication form | Allows users to continue with Google |   |
| **UploadButton** | Black filled button with upload icon | Workspace and Upload pages | Used to upload files |   |
| **NewChatButton** | Black filled button with plus icon | Chat page | Starts a new chat session |   |
| **NewSearchButton** | filled button with plus icon | Search page | Starts a new paper search |   |
| **ExportButton** | Gray/toolbar button with export icon | Citation editor | Exports article or citation content |   |
| **SendButton** | Icon button | Chat and search inputs | Sends chat messages or search queries |  |
| **ThemeToggleButton** | Small rounded icon button | Top-right header / Search page | Allows the user to switch between light mode and dark mode |  |
| **NotificationButton** | Small rounded icon button | Top-right header / Search page | Opens the notification dropdown |  |

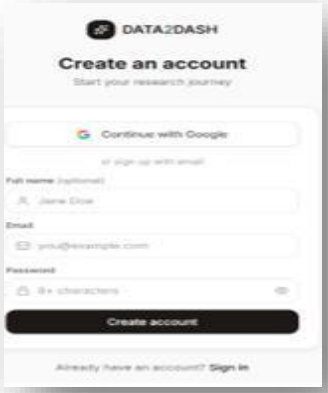
**2.7 Custom Components**

The frontend is built around reusable components such as navigation bars, agent cards, search result cards, upload panels, chat inputs, citation editors, and video generation panels. This improves consistency and makes the project easier to extend with new agents in the future

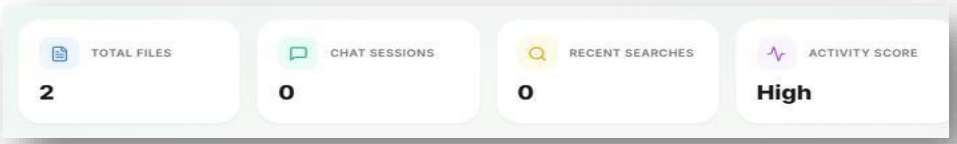
1. **1 Sidebar Component**
   Used across all main application screens. It displays the DATA2DASH logo, main navigation items, page action button, and user profile area.



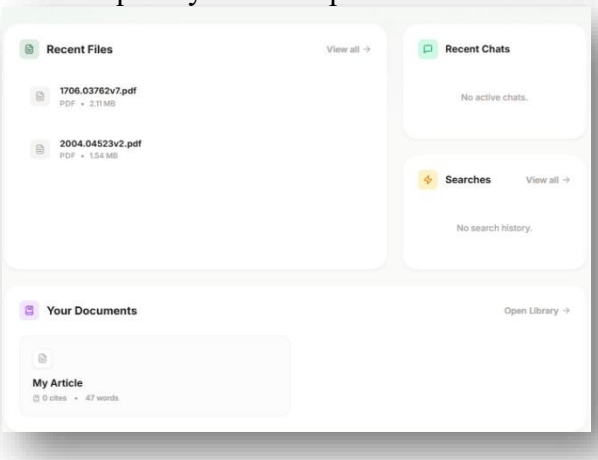

2. **Authentication Form Component**
   It contains the Google login button, email field, password field, and account creation fields



3. **Workspace Dashboard Cards**
   Used in the Workspace dashboard. These cards display important user activity data such as total files, chat sessions, recent searches, and activity score.

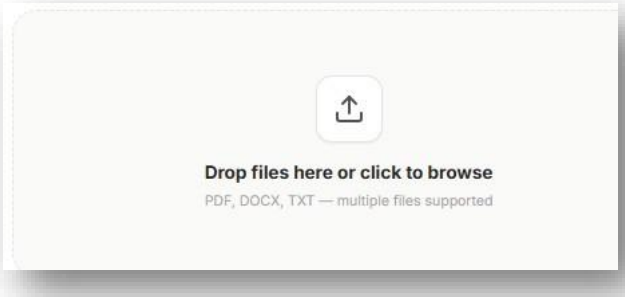



4. **Workspace Activity Panels**
   Used in the Workspace dashboard. These panels show recent files, recent chats, and search history, so the user can quickly return to previous work.

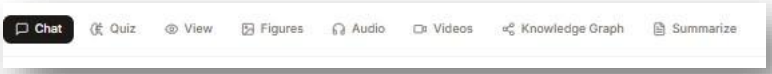



5. **Search Input and Suggestion Chips**
   Used in the Chat and Search pages. The search input allows the user to type research questions or paper search queries, while the suggestion chips provide quick topics such as RAG, Transformers, GAN, and BERT.

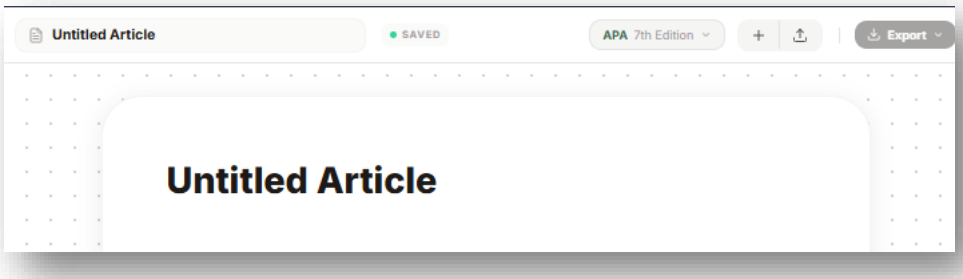
6. **Upload Dropzone Component**
   Used in the Upload page. It allows users to drag and drop files or browse files from their device. It supports document upload for analysis.




7. **Document Tabs Component**
   Used in the PDF analysis workspace. It provides access to document tools such as Chat, Quiz, View, Figures, Audio, Videos, Knowledge Graph, and Summarize.




8. **Citation Editor Component**
   Used in the Citation page. It provides an academic writing editor with citation format selection, export option, save status, and word/character count.




**2.8 AI Agents Displayed in the Frontend**

The DATA2DASH presentation lists nine intelligent agents, including Intelligent Search, Paper Summarizer, Knowledge Graph, Research Timeline, Multimodal RAG, AI Podcast Generator, Video Recommender, Citation Agent, and Critical Reviewer.
In the final implementation, the **YouTube Video Recommender** was updated to a more useful **Video Agent**. Instead of only recommending YouTube videos, the Video Agent generates a simple presentation about a topic and explains it using audio.

| Agent | Frontend Page / Component | Description |
|---|---|---|
| **Intelligent Search Agent** | Search Page | Finds relevant academic papers using semantic search |
| **Paper Summarizer** | Summary Component | Generates concise summaries of research papers |
| **Knowledge Graph Agent** | Visualization Page | Shows relationships between papers, concepts, and citations |
| **Research Timeline Agent** | Timeline Component | Shows the development of a research topic over time |
| **Multimodal RAG Agent** | Upload / PDF Chat Page | Explains text, equations, graphs, and tables from uploaded documents |
| **AI Podcast Generator** | Audio / Listen Page | Converts paper content into audio explanations |
| **Video Agent** | Video Agent Page | Generates a simple presentation about a topic and explains it using audio |
| **Citation Agent** | Citation Page | Generates APA, IEEE, and other citation formats |
| **Critical Reviewer** | Review Page | Reviews methodology, limitations, and paper quality |

### 2.10 Responsiveness

Since the frontend uses Tailwind CSS and a component-based React structure, the interface can be designed responsively across different screen sizes.

| Device Size | Responsive Behavior to Test / Show in Report |
|---|---|
| **Desktop** | Show the full dashboard layout with navigation, cards, side panels, and multi-column sections |
| **Tablet** | Show that cards and sections resize into fewer columns |
| **Mobile** | Show that components stack vertically, forms become full-width, and navigation is simplified |

### 2.11 Required Frontend Features

| Required Feature | How DATA2DASH Includes It |
|---|---|
| **Sign In / Sign Up** | Authentication screen with Google and email login |
| **Navigation** | Navbar links between Chat, Search, Upload, Citation, and other pages |
| **Notification** | Notifications can inform users when summaries, citations, presentations, or audio are ready |
| **Responsive Design** | Layout adapts to desktop, tablet, and mobile views |
| **Dynamic Content** | Search results, summaries, citations, uploaded documents, generated presentations, and audio outputs are loaded dynamically |

### 2.12 Final Frontend Summary

The frontend of DATA2DASH is implemented as a React and TypeScript web application using Vite and Tailwind CSS. The project follows a professional folder structure with separate folders for API logic, assets, reusable components, layouts, pages, styles, and TypeScript types. The interface uses a consistent design system based on dark brand colors, green accents, cream backgrounds, and DM Sans typography. Main screens include authentication, dashboard, chat, academic search, upload/PDF viewer, citation, notifications, and the updated Video Agent. The frontend connects users to multiple AI agents such as Intelligent Search, Paper Summarizer, Knowledge Graph, Multimodal RAG, Citation Agent, Critical Reviewer, AI Podcast Generator, and Video Agent. The Video Agent improves the

original YouTube recommender idea by generating a topic-based presentation and explaining it with audio, making the platform more interactive and useful for learning.

# Section 03: Backend

### 3.1 Backend Overview

The backend of Data2Dash AI is responsible for handling user authentication, file uploads, chat sessions, document processing, and database storage. The system uses a Python backend connected to a local SQLite database named data2dash.db. The backend receives requests from the frontend, processes them, stores the required data in the database, and sends dynamic responses back to the user interface.

The backend runs locally on:

http://localhost:8000

while the frontend runs on:

http://localhost:5173

This separation allows the frontend to focus on user interaction, while the backend manages logic, data storage, authentication, and AI-related operations.

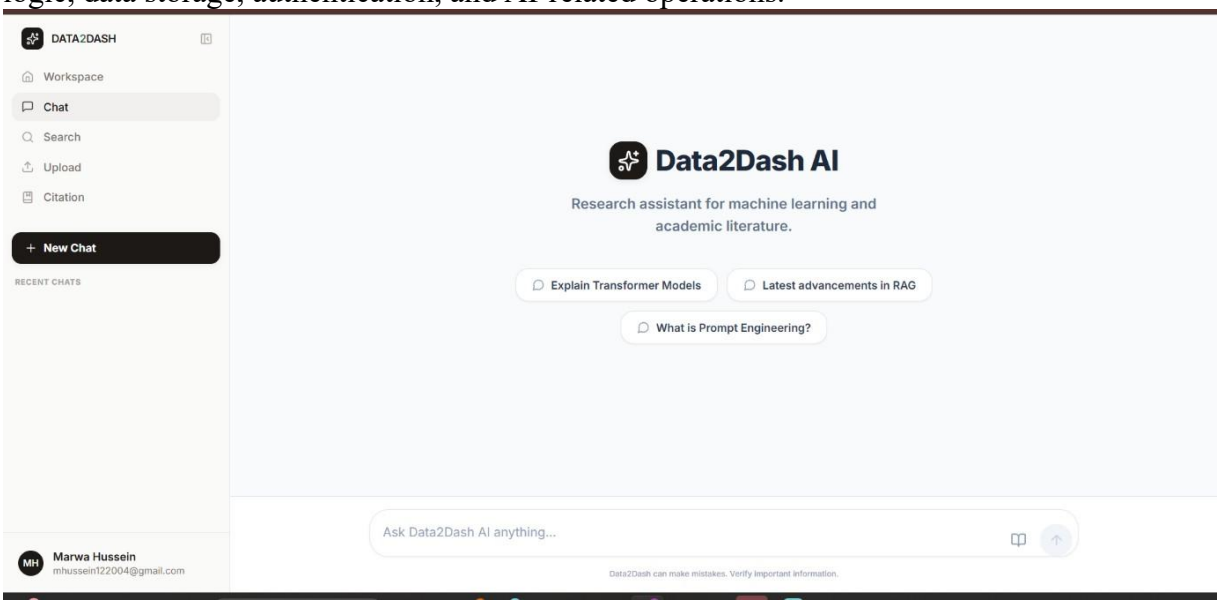



Figure 3.1: Data2Dash AI frontend connected to the backend service.

### 3.2 Backend Technologies

**The following table shows the main backend technologies used in the project.**

| Technology | Usage in the Backend |
|---|---|
| **Python** | **Main programming language used to build the backend logic.** |
| **FastAPI** | **Backend framework used to create API endpoints and handle frontend requests.** |
| **Uvicorn** | **ASGI server used to run the FastAPI backend locally.** |
| **SQLite** | **Local relational database used to store application records during development.** |
| **SQLAlchemy** | **Object Relational Mapper used to connect Python backend models with database tables.** |
| **DB Browser for SQLite** | **Tool used to inspect, test, and verify database records.** |
| **Python-dotenv** | **Used to load environment variables from the .env file.** |

| | |
|---|---|
| **JWT Authentication** | **Used to manage secure authenticated user sessions.** |
| **Password Hashing** | **Used to store user passwords securely as hashed values instead of plain text.** |
| **File Processing** | **Used to handle uploaded PDF files and store file metadata.** |

### 3.3 Database Structure

The application uses an SQLite database file named:
data2dash.db
The database file is located inside the backend folder:
**backend_python/data2dash.db**
The database contains multiple tables that support the main features of the application. Each table is responsible for storing a specific type of data.
**The main tables are:**

- **users**
- **workspaces**
- **files**
- **file_versions**
- **chat_sessions**
- **chat_messages**
- **search_history**
- **documents**

**The purpose of each table is described below:**

| Table Name | Purpose |
|---|---|
| **users** | **Stores user account information such as email, name, hashed password, and account status.** |
| **workspaces** | **Stores workspace information related to each user.** |
| **files** | **Stores uploaded file metadata such as filename, file type, size, and storage path.** |
| **file_versions** | **Stores versioning information for uploaded files if file versions are created.** |
| **chat_sessions** | **Stores chat session information such as title, user ID, workspace ID, and timestamps.** |
| **chat_messages** | **Stores the actual user and AI messages inside each chat session.** |
| **search_history** | **Stores user search queries and related metadata.** |
| **documents** | **Stores generated or processed document content, citations, style, and word count.** |

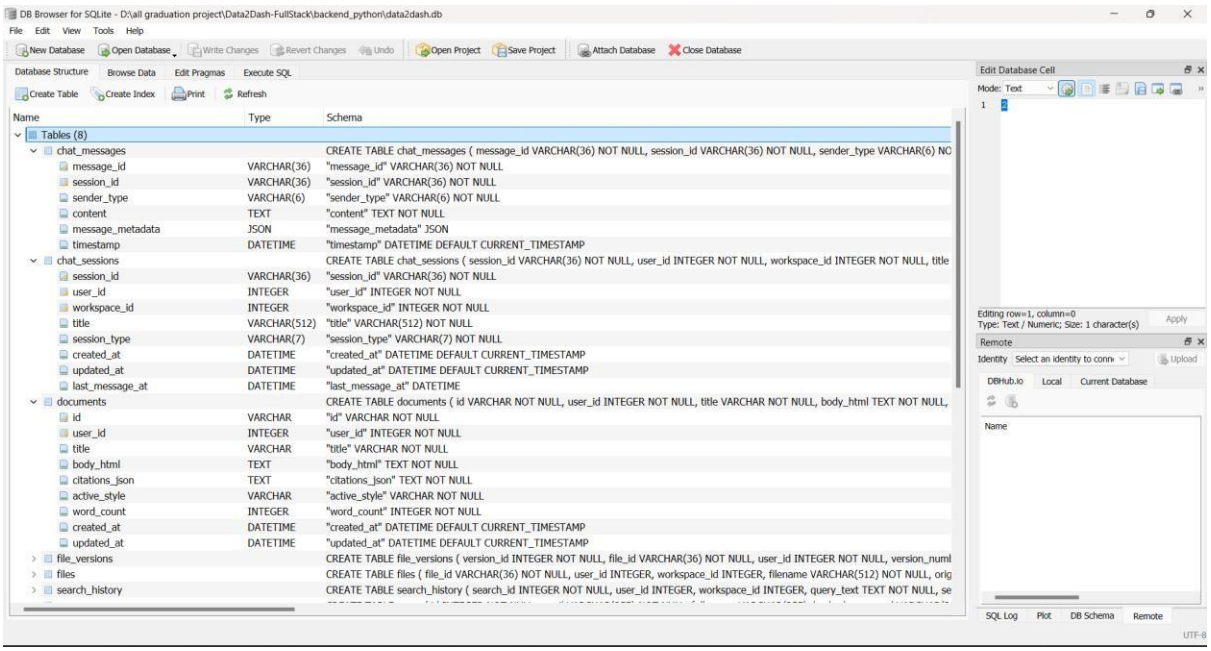


Figure 3.2: Database schema showing the main SQLite tables used by the backend.

---

### 3.4 User Authentication and User Records

The application includes a sign-up and sign-in system. When a new user creates an account from the frontend, the backend receives the submitted data and stores a new record in the users table.

**The users table contains fields such as:**

- **id**
- **email**
- **full_name**
- **hashed_password**
- **google_id**
- **avatar_url**
- **is_active**
- **created_at**
- **updated_at**
- **last_login_at**

The password is not stored as plain text. Instead, the backend stores it as a hashed value in the hashed_password column. This improves security because the original password is not directly visible in the database.

During testing, two user accounts were created successfully. After creating the accounts from the frontend, the records appeared in the users table inside the SQLite database. This proves that the registration form is connected to the backend and database correctly.

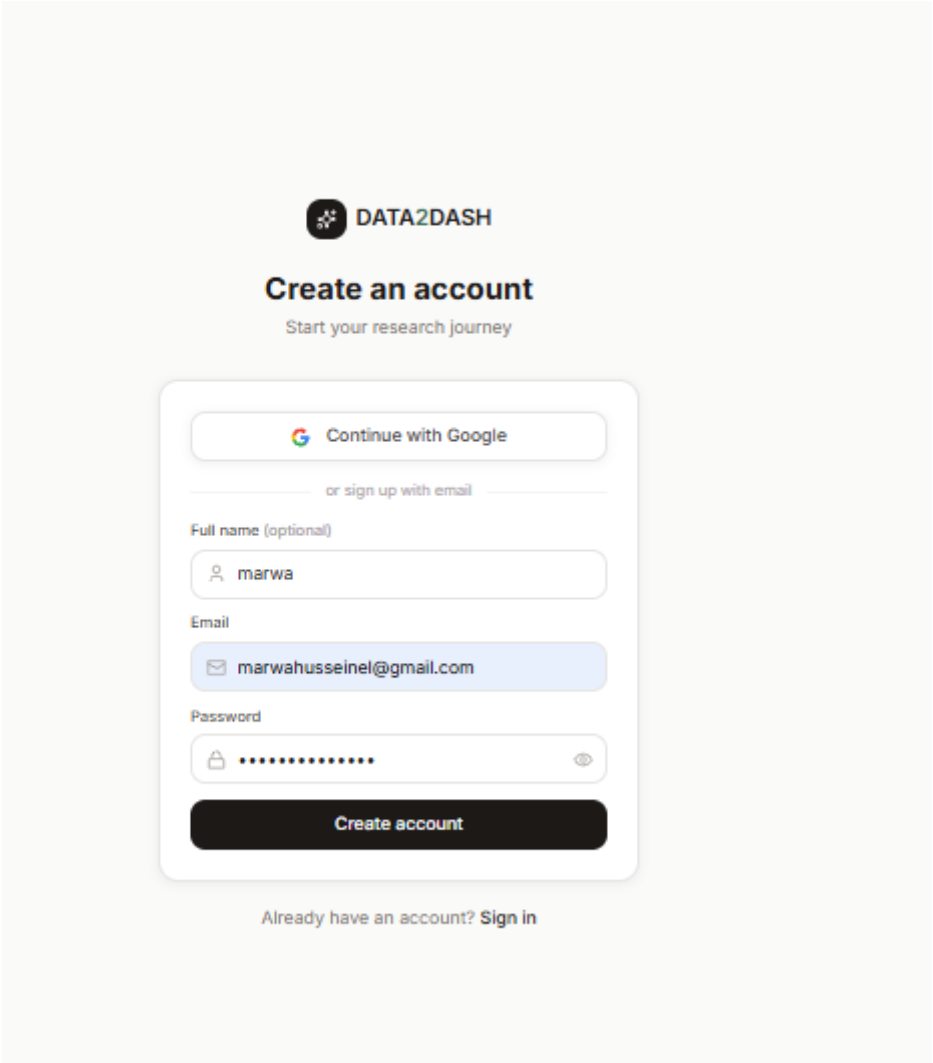
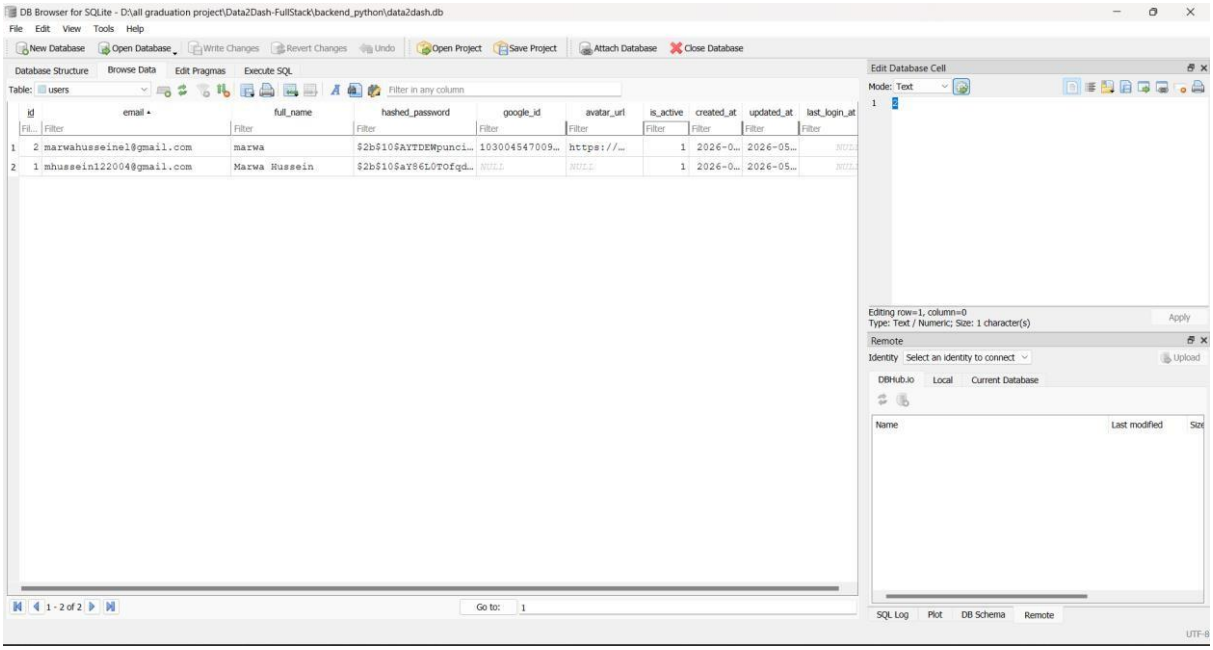



**Figure 3.3: User registration form filled with test account data before submitting it to the backend.**




**Caption:**
**Figure 3.4: User records stored in the SQLite database after account creation.**

**Explanation:**
The records in the users table confirm that the backend successfully receives account data from the frontend and saves it in the database. The hashed_password field shows that passwords are stored securely as hashed values instead of plain text.

### 3.5 Workspaces

Each user has a workspace that organizes their activity inside the application. The workspaces table links users with their related workspace data.
The workspaces table includes fields such as:

| Column Name | Description |
|---|---|
| **id** | **Unique identifier for each workspace record.** |
| **user_id** | **References the user who owns this workspace.** |
| **name** | **Stores the workspace name.** |
| **created_at** | **Stores the date and time when the workspace was created.** |

During testing, the SQL count query showed that two workspace records were created. This means that when users are created or logged in, workspace-related data is also stored in the database.
This helps the backend organize user-specific files, chats, and documents under the correct user workspace.

---

### 3.6 Chat Sessions and Chat Messages

The chat feature is one of the main dynamic parts of the Data2Dash AI application. When the user sends a question from the frontend, the backend receives the request, creates or updates a chat session, stores the conversation in the database, and returns the AI response to be displayed in the chat interface.
The chat system mainly depends on two database tables: chat_sessions and chat_messages.

**Chat Sessions Table**

The chat_sessions table stores general information about each conversation. Each chat session belongs to a specific user and workspace.

**Table 3.4: Structure of the chat_sessions table**

| Column Name | Description |
|---|---|
| **session_id** | **Unique identifier for each chat session.** |
| **user_id** | **References the user who created the chat session.** |
| **workspace_id** | **References the workspace related to the chat session.** |
| **title** | **Stores the generated or selected title of the conversation.** |
| **session_type** | **Defines the type of chat session.** |
| **created_at** | **Stores the date and time when the chat session was created.** |
| **updated_at** | **Stores the date and time when the chat session was last updated.** |

| | |
|---|---|
| **last_message_at** | **Stores the date and time of the last message in the session.** |

**Chat Messages Table**

The chat_messages table stores the actual messages inside each chat session. It includes both the user messages and the AI responses. This allows the application to retrieve and display previous conversations dynamically.

**Table 3.5: Structure of the chat_messages table**

| Column Name | Description |
|---|---|
| **message_id** | **Unique identifier for each message.** |
| **session_id** | **References the chat session that the message belongs to.** |
| **sender_type** | **Identifies whether the message was sent by the user or by the AI assistant.** |
| **content** | **Stores the text content of the message.** |
| **message_metadata** | **Stores additional message-related information if available.** |
| **timestamp** | **Stores the date and time when the message was created.** |

For example, during testing, the user asked:

**Explain Transformer Models**

After submitting this question, the backend created a new chat session and stored the related messages in the database. The chat session also appeared dynamically in the frontend sidebar under Recent Chats.

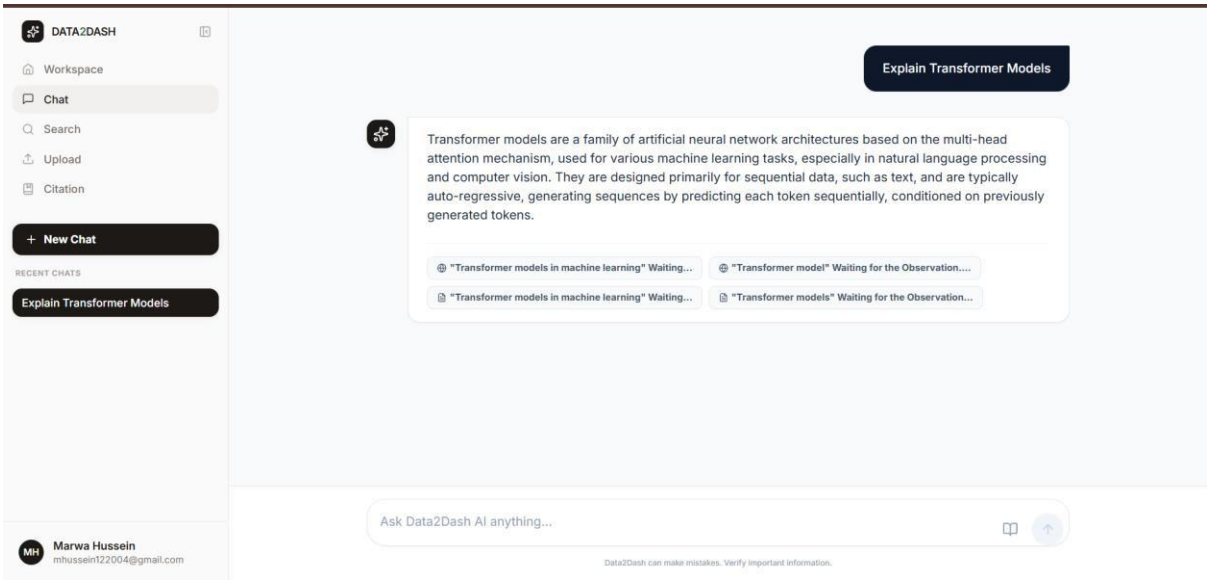



**Figure 3.5: Dynamic AI chat response rendered in the frontend.**

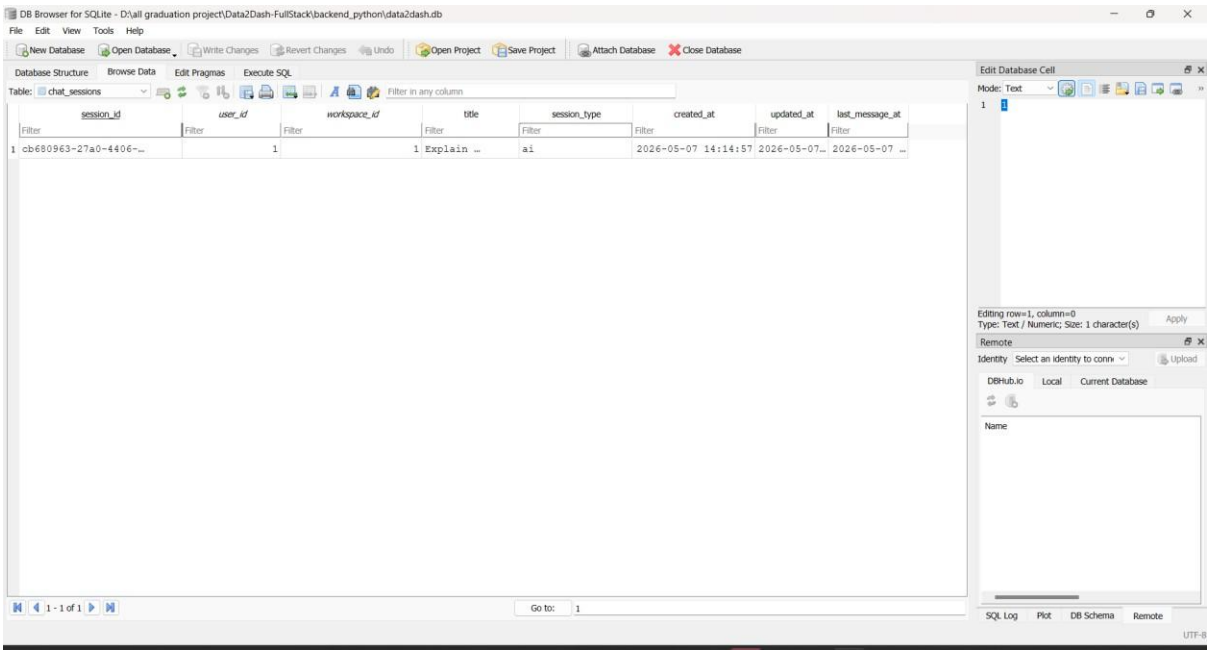



**Figure 3.6: Chat sessions saved in the database after user interaction.**

The user also tested a follow-up question, such as:

explain in more details multi-head attention mechanism

This follow-up interaction proves that the chat feature supports continuous conversation and that the frontend updates dynamically based on user input and backend responses.

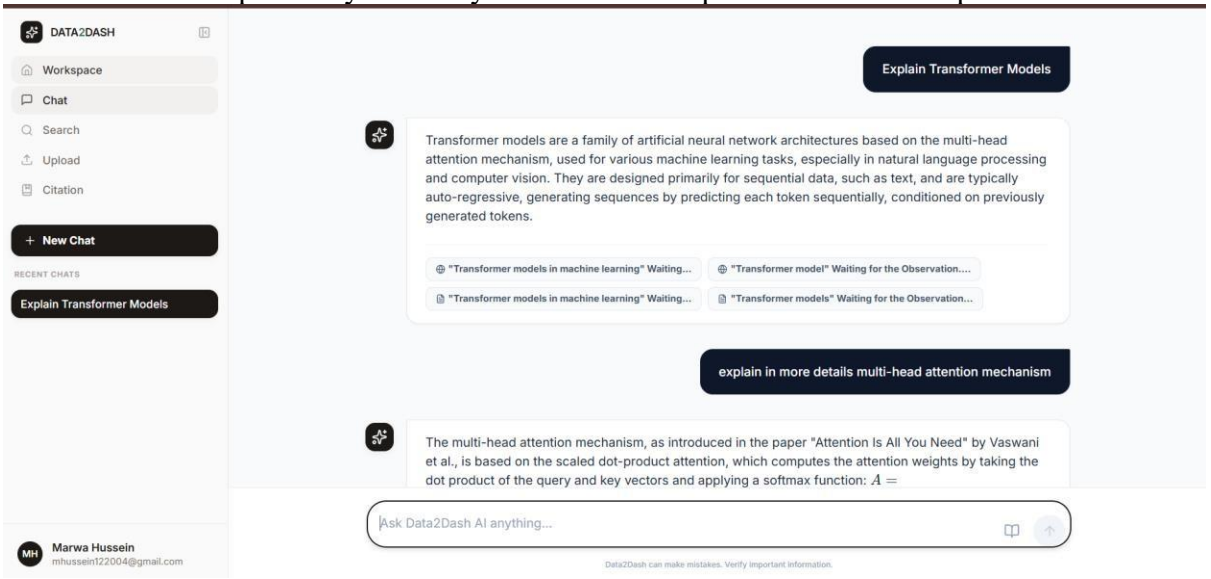



Figure 3.7: Follow-up chat interaction rendered dynamically in the application.

**Explanation**

The chat screenshots and database records prove that the chat content is dynamic. When the user sends a message, the frontend displays the response in the chat interface, while the backend stores the session and messages in the database. This makes it possible for the application to show previous conversations again in the sidebar and maintain chat history for each user.

This confirms that the chat feature is not static; it is connected to backend database records through the chat_sessions and chat_messages tables.

### 3.7 File Upload and File Records

The backend handles file uploads by receiving files from the frontend, saving their metadata, and storing the related information in the SQLite database. When a user uploads a PDF file, the backend inserts a new record into the files table.

**Table 3.6: Structure of the files Table**

| Column Name | Description |
|---|---|
| file_id | Unique identifier for each uploaded file. |
| user_id | References the user who uploaded the file. |
| workspace_id | References the workspace where the file belongs. |
| filename | Stores the system-generated file name. |
| original_name | Stores the original name of the uploaded file. |
| file_type | Stores the type of the uploaded file, such as PDF. |
| mime_type | Stores the MIME type of the file, such as application/pdf. |
| size_bytes | Stores the file size in bytes. |
| storage_path | Stores the path where the uploaded file is saved. |

During testing, a PDF file named: 1810.04805v2.pdf
was uploaded successfully. The file appeared in the frontend upload page, and the backend inserted a new record into the files table.

**Table 3.7: File Upload Test Result**

| Test Item | Result |
|---|---|
| Tested Feature | PDF file upload |
| Uploaded File Name | 1810.04805v2.pdf |
| Frontend Result | The uploaded PDF appeared inside the Data2Dash AI interface. |
| Backend Result | File metadata was inserted into the files table. |
| Database Evidence | A new record appeared in DB Browser inside the files table. |

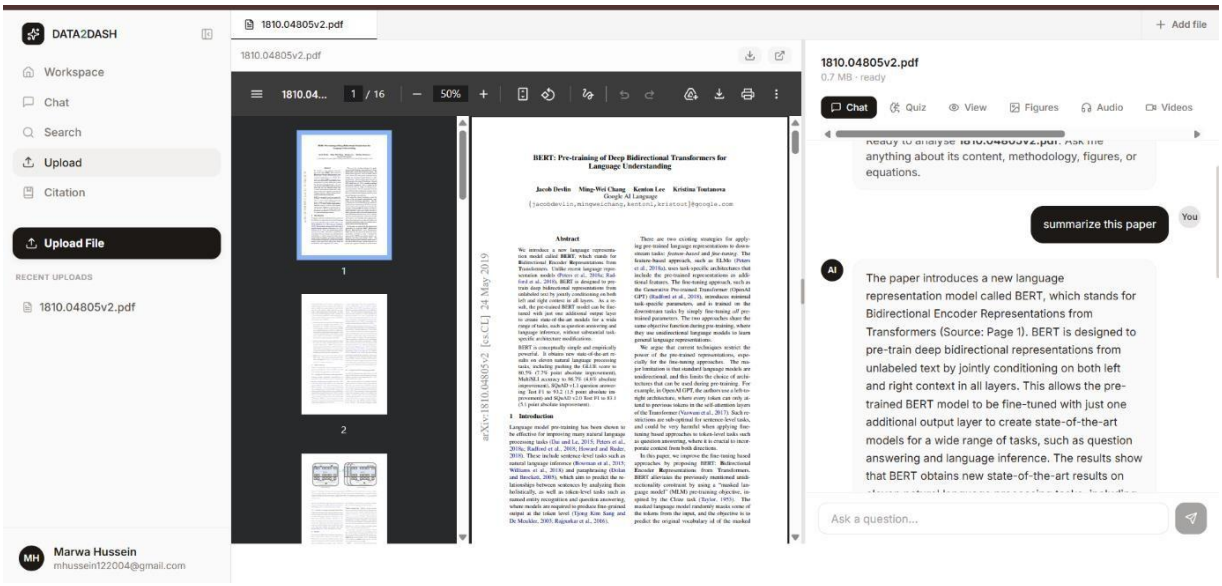



Figure 3.8: Uploaded PDF file displayed inside the Data2Dash AI interface.

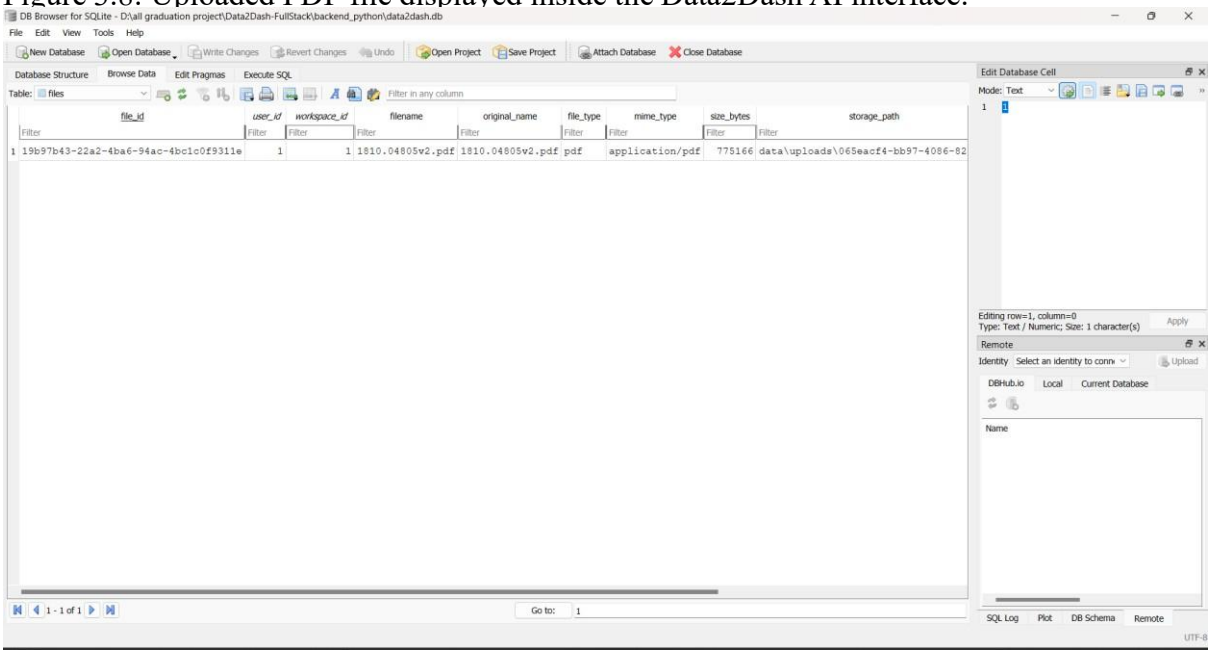



Figure 3.9: Uploaded file metadata stored in the files table.

The file upload test confirms that the backend stores uploaded file metadata correctly. The frontend displays the uploaded PDF, while the database stores the file information for future retrieval.

### 3.8 Document Records

The documents table stores document-related data created or processed by the application. This table can store generated document content, citation information, formatting style, word count, and timestamps.

**Table 3.8: Structure of the documents Table**

| Column Name | Description |
|---|---|
| **id** | **Unique identifier for each document record.** |
| **user_id** | **References the user who owns or created the document.** |

| | |
|---|---|
| **title** | **Stores the document title.** |
| **body_html** | **Stores the document body content in HTML format.** |
| **citations_json** | **Stores citation data in JSON format.** |
| **active_style** | **Stores the selected formatting or citation style.** |
| **word_count** | **Stores the number of words in the document.** |
| **created_at** | **Stores the date and time when the document was created.** |
| **updated_at** | **Stores the date and time when the document was last updated.** |

During testing, the documents table contained two records with the title: **My Article**

**Table 3.9: Document Records Test Result**

| **Test Item** | **Result** |
|---|---|
| **Tested Table** | **documents** |
| **Number of Records** | **2** |
| **Example Document Title** | **My Article** |
| **Backend Result** | **The backend stored document-related data in the database.** |
| **Database Evidence** | **Two document records appeared in DB Browser.** |

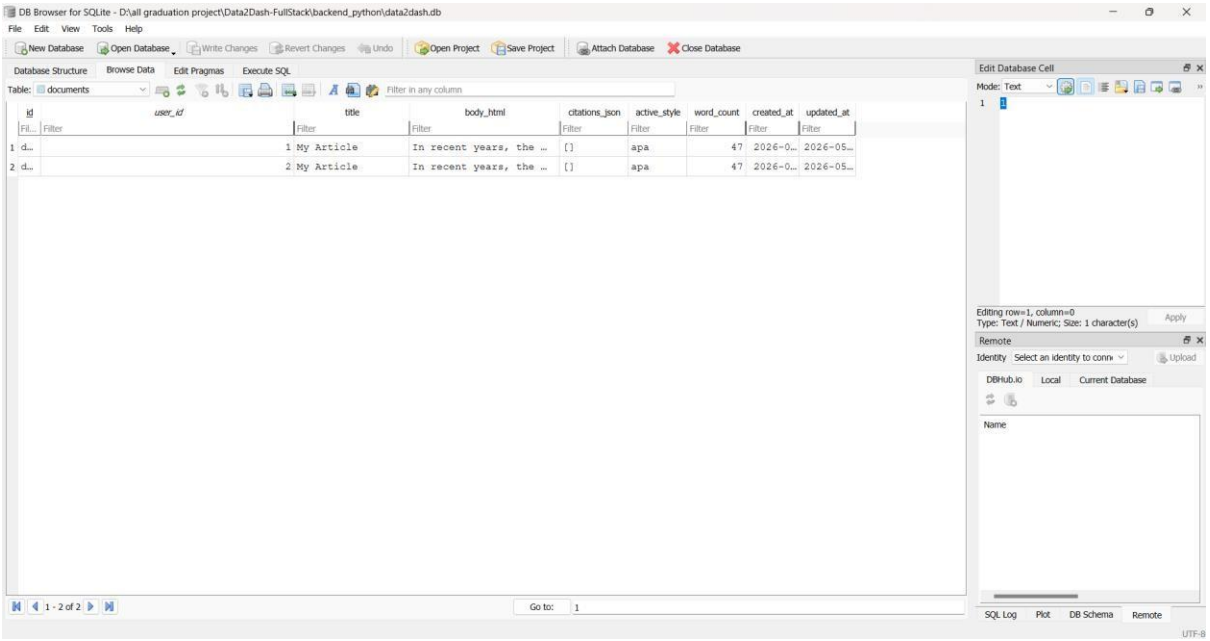



**Figure 3.10: Document records stored in the SQLite database.**

The document records show that the backend supports storing structured document content. This content can later be retrieved and displayed dynamically in the frontend

### 3.9 Database Testing Using SQL Query

To verify backend database integration, an SQL query was executed using DB Browser for SQLite. The purpose of this query was to count the number of records in each main database table after testing the application features.

**Table 3.10: SQL Query Purpose**

| Item | Description |
|---|---|
| **Tool Used** | **DB Browser for SQLite** |
| **Tested Database** | **data2dash.db** |
| **Query Purpose** | **Count the number of records in each main database table.** |
| **Reason** | **To prove that frontend actions were stored in the backend database.** |

**The query used was:**

```
SELECT 'users' AS table_name, COUNT(*) AS rows_count FROM users
UNION ALL
SELECT 'workspaces', COUNT(*) FROM workspaces
UNION ALL
SELECT 'files', COUNT(*) FROM files
UNION ALL
SELECT 'file_versions', COUNT(*) FROM file_versions
UNION ALL
SELECT 'chat_sessions', COUNT(*) FROM chat_sessions
UNION ALL
SELECT 'chat_messages', COUNT(*) FROM chat_messages
UNION ALL
SELECT 'search_history', COUNT(*) FROM search_history
UNION ALL
SELECT 'documents', COUNT(*) FROM documents;
```

**Table 3.11: Database Record Count Result**

| Table Name | Number of Records | Explanation |
|---|---|---|
| **users** | **2** | **Two user accounts were created during testing.** |
| **workspaces** | **2** | **Two workspace records were created for the users.** |
| **files** | **1** | **One PDF file was uploaded and stored.** |
| **file_versions** | **0** | **No file versioning operation was performed during this test.** |
| **chat_sessions** | **3** | **Three chat sessions were created.** |
| **chat_messages** | **8** | **Eight chat messages were stored, including user messages and AI responses.** |
| **search_history** | **0** | **The search feature was not tested or not recorded during this specific test.** |

| | | |
|---|---|---|
| **documents** | **2** | **Two document records were stored in the database.** |

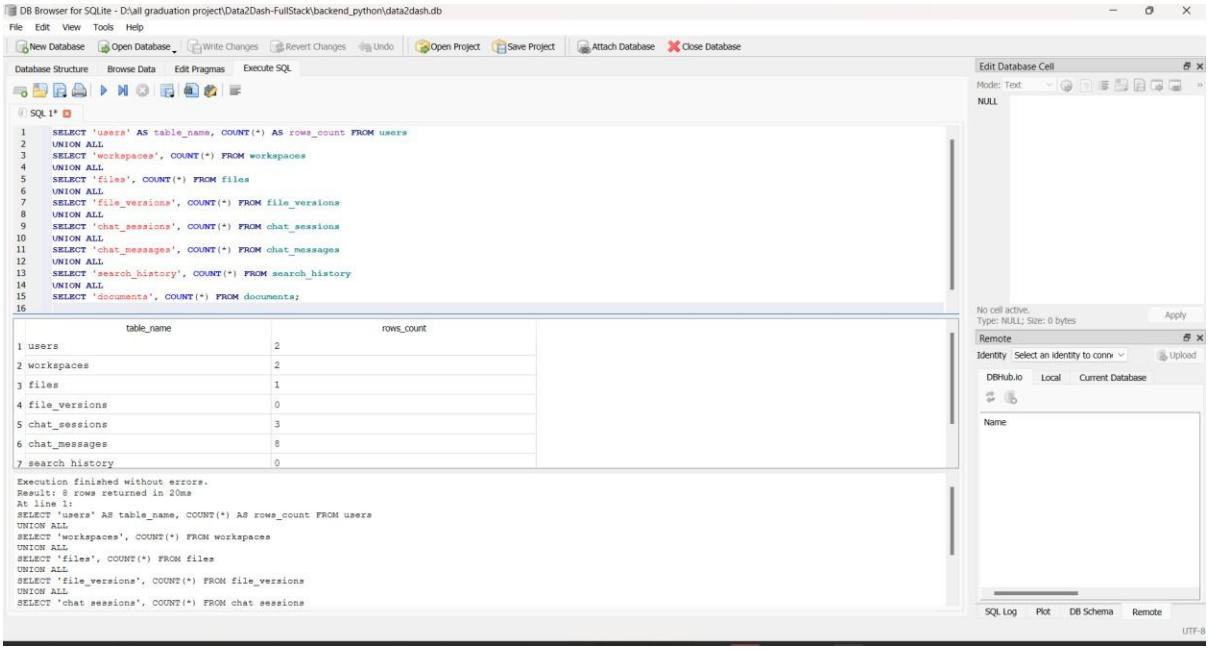



**Figure 3.11: SQL query result showing record counts for each backend database table.**
The SQL count result provides clear evidence that the backend is storing data in multiple database tables. It confirms that user registration, workspace creation, chat sessions, chat messages, file uploads, and document storage were successfully tested.

### 3.10 Dynamic Content Rendered from the Database

Several parts of the frontend are dynamic and depend on backend database records. This means the application does not only display static content. Instead, the interface changes according to the data stored and retrieved from the database.

**Table 3.12: Examples of Dynamic Content Rendered from the Database**

| Dynamic Content | Related Database Table | Explanation |
|---|---|---|
| **Logged-in user name and email** | **users** | **The sidebar displays the logged-in user's name and email based on stored user data.** |
| **Recent chat titles** | **chat_sessions** | **The sidebar displays recent chat titles retrieved from saved chat sessions.** |
| **Chat messages and AI responses** | **chat_messages** | **The chat interface displays stored user messages and AI responses.** |
| **Uploaded file list** | **files** | **Uploaded files are displayed based on file metadata stored in the database.** |

| | | |
|---|---|---|
| **Document records** | **documents** | **Stored documents can be retrieved and displayed dynamically.** |
| **Workspace-related data** | **workspaces** | **User activity is organized under workspace records.** |

For example, after the user logs in, the sidebar displays the user name and email. These values are related to the stored user record. Also, when the user creates a new chat, the chat title appears under Recent Chats. This information is stored in the chat_sessions table and rendered dynamically in the frontend.
When the user uploads a file, the uploaded file appears in the upload section. The backend stores its metadata in the files table, and the frontend displays it to the user.

**Table 3.13: Dynamic Content Evidence**

| Frontend Area | What Appears in the UI | Backend Evidence |
|---|---|---|
| **Sidebar** | **User name and email** | **Stored in the users table.** |
| **Recent Chats** | **Chat titles such as "Explain Transformer Models"** | **Stored in the chat_sessions table.** |
| **Chat Interface** | **User questions and AI responses** | **Stored in the chat_messages table.** |
| **Upload Page** | **Uploaded PDF file** | **Stored in the files table.** |
| **Documents Section** | **Document title and content** | **Stored in the documents table.** |

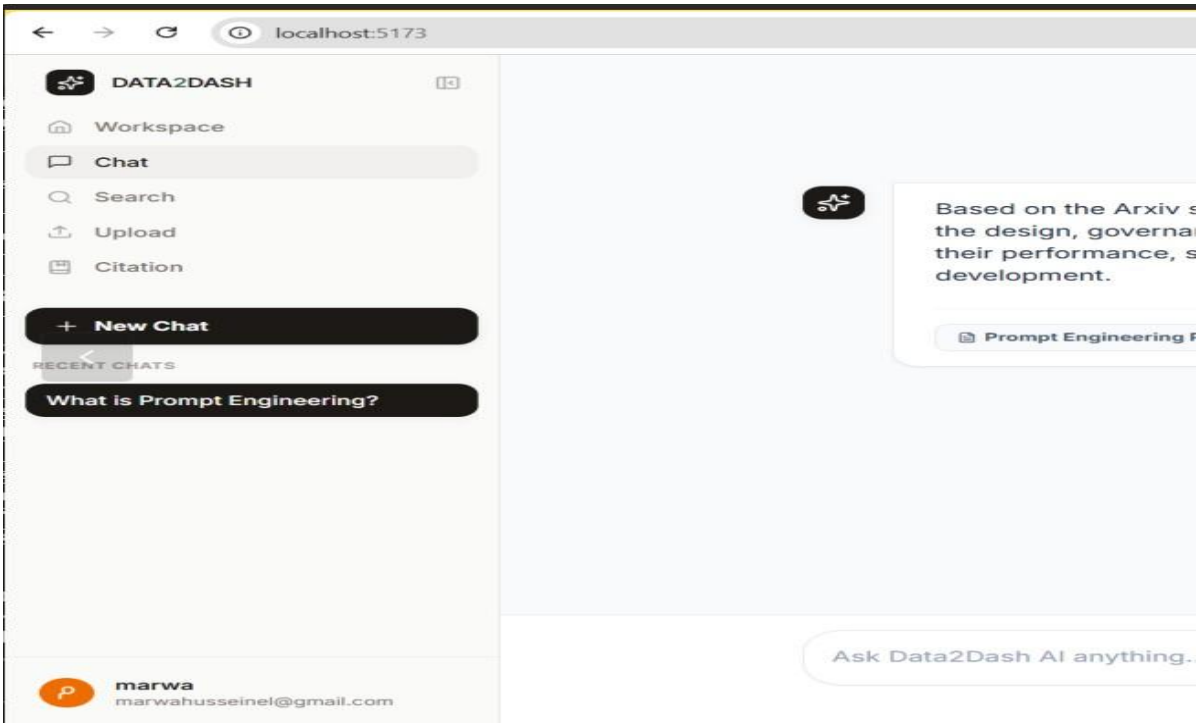



**Figure 3.12: Dynamic user profile and recent chat content rendered from backend data.**
This figure proves that the frontend is not only displaying static content. The sidebar changes based on the logged-in user and the user's saved chat sessions. This confirms that dynamic content is being retrieved from backend data and displayed in the user interface.

| Required Element | Should Be Visible? |
|---|---|
| User name at the bottom of the sidebar | Yes |
| User email | Yes |
| Recent Chats list | Yes |
| A chat title such as "Explain Transformer Models" or "What is Prompt Engineering?" | Yes |

### 3.11 Evidence of Backend and Database Integration

The testing results prove that the frontend, backend, and database are connected successfully.

**Table 3.14: Backend and Database Integration Evidence**

| Test Action | Frontend Result | Database Result |
|---|---|---|
| Create account | User account was created and displayed in the sidebar. | A new record was inserted into the users table. |
| Start chat | Chat response was displayed in the chat interface. | A new record was inserted into the chat_sessions table. |
| Send messages | User and AI messages were displayed in chat. | Messages were stored in the chat_messages table. |
| Upload PDF | Uploaded PDF appeared in the upload page. | File metadata was stored in the files table. |
| Store document content | Document data was available in the system. | Records were stored in the documents table. |
| Run SQL count query | Database records were summarized. | Row counts were displayed for each table. |

This confirms that the application does not rely on static data only. The content changes based on user actions, and these actions are reflected in the backend database.

### 3.12 Backend Testing Summary

The backend was tested by performing real actions from the frontend and then checking the database records using DB Browser for SQLite.

**Table 3.15: Backend Testing Steps**

| Step Number | Testing Step | Verification Method |
|---|---|---|
| 1 | Created user accounts from the frontend. | Checked the users table in DB Browser. |
| 2 | Verified that users were inserted into the database. | Confirmed user records appeared in the users table. |
| 3 | Sent chat questions from the frontend. | Checked the chat interface and database records. |
| 4 | Verified that chat sessions were inserted. | Checked the chat_sessions table. |
| 5 | Verified that chat messages were inserted. | Checked the chat_messages table. |
| 6 | Uploaded a PDF file. | Checked the upload page in the frontend. |

| | | |
|---|---|---|
| **7** | **Verified that file metadata was inserted.** | **Checked the files table.** |
| **8** | **Checked generated document records.** | **Checked the documents table.** |
| **9** | **Ran an SQL query to count records.** | **Used DB Browser Execute SQL tab.** |

**Table 3.16: Backend Testing Summary Result**

| Tested Area | Status | Evidence |
|---|---|---|
| **User authentication** | **Passed** | **User records appeared in the users table.** |
| **Workspace creation** | **Passed** | **Workspace records appeared in the workspaces table.** |
| **Chat sessions** | **Passed** | **Chat sessions appeared in the chat_sessions table.** |
| **Chat messages** | **Passed** | **Messages appeared in the chat_messages table.** |
| **File upload** | **Passed** | **Uploaded PDF metadata appeared in the files table.** |
| **Document records** | **Passed** | **Document records appeared in the documents table.** |
| **SQL database verification** | **Passed** | **Record count query returned results for all main tables.** |

The results confirm that the backend is working correctly. It receives data from the frontend, stores it in the SQLite database, retrieves dynamic content, and supports the main features of the application.

## Conclusion

In conclusion, Data2Dash AI is a functional web-based research assistant that helps users upload academic papers, interact with documents, ask research questions, and organize their research work. The project followed a complete development process including UX research, problem definition, wireframing, frontend design, backend implementation, and database testing.

The frontend provides a responsive and organized interface with clear navigation between the main features, while the backend manages authentication, file storage, chat sessions, messages, documents, and user data. The SQLite database testing proved that the application stores real user actions and displays dynamic content from the backend.

Overall, the project successfully meets its main goal by connecting the frontend, backend, and database into one working system. The application can be further improved in the future by adding more advanced citation tools, better search history, stronger document analysis, and online deployment.